



Hallucinations in generative artificial intelligence and large language models: tests, datasets, detection and correction methods

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Abstract

Generative Artificial Intelligence (GAI) and Large Language Models (LLMs) have demonstrated significant capabilities in generating human-like content; however, they exhibit a propensity to fabricate spurious information, a phenomenon often termed *hallucination*. This review paper provides a comprehensive overview of hallucinations in GAI and LLMs. More specifically, this review encompasses their definitions, underlying mechanisms, taxonomies, commonly used tests, and datasets for evaluating hallucinations. In addition, this review dives into intrinsic and extrinsic factors contributing to these inaccuracies, including limitations in model architectures, training data biases, and inference algorithms, as well as examines various detection strategies [e.g., post-hoc consistency checks, external fact-checking, contrastive learning, uncertainty calibration methods, and Retrieval-Augmented Generation (RAG)]. The review also synthesizes a range of correction and mitigation techniques, from proactive measures during training to hybrid approaches that combine detection and intervention. Finally, this review integrates qualitative assessments and comparative insights to delineate the impact of hallucinations on user trust and acceptability, and to shed light on current challenges and future research trends.

Keywords Review · Large language models (LLMs) · Generative AI (GAI)

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1 Introduction

Generative artificial intelligence (GAI) has rapidly evolved from a niche research area into a transformative domain (Lv, 2023). Examples of GAI systems include diffusion and large language models (LLMs). Such systems often host multi-billion-parameter architectures with sophisticated attention/self-supervised mechanisms capable of generating human-like output (e.g., images, text, etc.) (Ooi et al., 2025). For example, LLMs can capture nuanced syntactic and semantic relationships to construct contextually coherent sentences, paragraphs, and dialogues (Gromova et al., 2023). The rise of such systems and other variants with generative multimodal capabilities has led to a surge in commercial interest (e.g., search engines, digital assistants, etc.) (OpenAI et al., 2023).

Despite these achievements, generative models sometimes invent fabricated information or *hallucinations* (Ji et al., 2025; Schaik et al., 2024). This term often describes outputs that appear syntactically correct or visually plausible but lack grounding in factual, consistent, or verifiable data (Abbasi et al., 2024). Such inaccuracies can be inconsequential in low-stakes settings, but they become problematic when individuals or organizations rely on the generated content in decision making (Pak et al., 2024). For example, misleading outputs in a particular context (e.g., medical diagnosis) could misinform decision-making (e.g., treatment) and hence might jeopardize patient well-being. In other instances, LLMs have been reported to confidently assert false legal precedents and scholarly citations (Alsbrook, 2024; Latif, 2025). This has not only caused user confusion but also sparked public debates about the trustworthiness of GAI (Zhang et al., 2024a). Although such instances might be resolved through user reports or developer interventions, the damage to public confidence often remains (Chen et al., 2023a; Coeckelbergh, 2025).

It is worth remembering that the growth of diffusion and LLMs has not only scaled up in terms of the number of used parameters but has also led to drastic shifts in performance characteristics (Rostam et al., 2024; Yang et al., 2024). While early versions of such systems exhibited noticeable failings when presented with complex reasoning tasks (Kasneci et al., 2023), subsequent versions are trained on much larger datasets and equipped with refined training procedures, and hence generally exhibit improved consistency. Yet, these enhancements did not eliminate hallucinations. Instead, they change how errors appear and make them more difficult to detect (Puccetti et al., 2024; Zhou et al., 2024a). As these technologies rise to mainstream applications, the stakes of misinformation increase, which in turn raises questions about how to maintain reliability, transparency, and accountability (Min et al., 2023). Therefore, and as one can expect, there is an urgent need to examine how GAI models produce, propagate, and potentially intensify erroneous or fabricated information.

Surprisingly, hallucinations are not a new/recent phenomenon, but rather, researchers have long recognized that generative models can produce outputs that deviate from factual reality (Huang, 2025). Early accounts of this phenomenon emerged in machine translation, where systems occasionally introduced content absent from the source text (Xu et al., 2023). These fabrications were labeled *hallucinated translations* to emphasize that the model's output was erroneous/imaginatively inserted. Follow-up studies then noted how, this issue was not confined to translation: sum-

marization models also exhibited the same tendency when dealing with lengthy or technical documents (Hamid, 2024; Perković et al., 2024; Reddy et al., 2024). A root cause identified in this research direction is that conventional training processes rely on maximizing likelihood without explicit truth constraints, leading models to over-rely on correlation patterns rather than genuine understanding (Sriramanan et al., 2024).

Subsequent studies focused on quantifying this tendency and analyzing why modern neural architectures still fall short of ensuring authenticity (Arteaga, 2024; Chen et al., 2024a). One line of inquiry proposed that inconsistencies in token distributions across diverse training data could cause internal embeddings to conflate semantically unrelated terms (Huang, 2025). As a result of this behavior, the model's attention mechanisms occasionally align on spurious cues (which give rise to fabricated statements/descriptions). Another influential perspective highlighted how the absence of systematic fact-checking during model training and optimization compounds the risk of hallucinated content (Augenstein et al., 2024). What is interesting, and concerning at the same time, to note is that whereas earlier language models might produce awkward/grammatically incorrect sentences when failing, more advanced systems generate fluent statements that subtly misstate factual details (Hu et al., 2024a). This capacity for plausible but inaccurate composition stresses the challenge of diagnosing hallucinations.

The drive to address hallucinations led to growing interest in alignment protocols to align model outputs with established bodies of knowledge (Liu et al., 2023; Wang et al., 2023a). Initial attempts focused on constraining generative processes through architectures that included a generative component that proposed a draft text, which a secondary verifier would then evaluate. If the verifier flagged inconsistencies, the system would regenerate or adjust the suspect segments (Ji et al., 2024). Though these early architectures demonstrated value, they often faced scalability obstacles for tasks involving large contexts. Thus, re-ranking techniques were used in response, in which multiple candidate outputs are generated and the sequence with the fewest anomalies is selected (Carraro, 2024). Studies employing this approach showed modest reductions in misstatements, but performance was heavily influenced by how effectively the re-ranker assessed factual consistency (which itself can be error-prone) (Meng et al., 2024).

Parallel efforts found that if a model's training data included contradictory statements or if it covered a given topic superficially, the learned representations became unreliable for complex queries (Waldo, 2024). This concern was echoed in specialized fields (e.g., medicine or engineering) where slight inaccuracies could carry intellectual or ethical implications. To address these deficits, some studies proposed domain-specific pretraining/finetuning on rigorously curated datasets (Anisuzzaman et al., 2025). Although that strategy improved performance, it proved demanding to maintain currency in rapidly changing fields, as the curated datasets required continuous updates. This limitation prompted further developments in incremental learning to enable models to ingest new information without extensive retraining (Perković et al., 2024). Similarly, the literature explores knowledge grounding in relation to hallucination (Addlesee, 2024). In knowledge-grounded dialogue systems, a GAI model derives responses from a set of retrieved documents or structured records rather than

relying solely on internal parameters. Empirical results indicated that retrieval-based approaches decreased the incidence of incorrect content (provided the retrieval component is accurate) (Huang & Huang, 2024). However, as mentioned above, the reliance on third-party or domain-specific knowledge bases posed logistical concerns in terms of how rapidly evolving knowledge often demands frequent re-indexing.

A closer look at generation methods indicated that certain inference strategies might amplify hallucinations (Barros et al., 2025). For example, beam search is considered a commonly used default approach in sequence-to-sequence frameworks. This approach has been shown to prioritize sequences that appear coherent while neglecting fidelity to factual constraints (Du et al., 2024). On the other hand, temperature sampling introduced random tokens that undermined clarity and noted how adjusting inference hyperparameters, such as limiting the search space or enforcing stricter coverage of source tokens, could mitigate fabricated content to an extent (Farquhar et al., 2024). However, such tunings may reduce the diversity or creativity of model outputs (Peeperkorn et al., 2024). This tension drove interest in selective decoding, in which the system identifies critical segments and applies stricter thresholds while allowing more flexibility in descriptive sections. Recent studies also look at how developers can systematically track hallucination rates over time (Jesson et al., 2024). This approach compares different versions of the same GAI system by measuring improvement or regression in factual alignment against a stable test suite of carefully curated prompts that expose known pitfalls (i.e., enabling tracking of whether newly introduced training data or architecture tweaks have a measurable impact) (Chen et al., 2024b; Lin et al., 2024).

Outside text-based GAI, related concerns of hallucination also extend to image synthesis and other modalities (Gao et al., 2024). Similar to their text counterparts, earlier generative image-based models produced disjointed visuals that were easy to detect. However, modern diffusion and transformer-based image generators tend to produce intricately detailed pictures (Bengesi et al., 2024). The line between a minor compositional error and a genuinely misleading scene can be blurry, as inaccurate object relationships/improbable scene configurations could pass unnoticed if the overall output appears polished (Stefaniia, 2024). This observation led to the development of multimodal consistency checks, in which a separate model compares the synthesized imagery with textual captions or ground-truth metadata.

Several surveys have examined hallucinations in LLMs in recent years (e.g., Huang et al. 2025; Alansari and Luqman 2025; Rawte et al. 2023a). These contributions have advanced the field by cataloging causes, detection methods, and mitigation strategies, though they have predominantly focused on text-based LLMs and have generally adopted binary or narrow taxonomic schemes (e.g., factuality vs. faithfulness). The present review departs from this baseline in several respects. First, its scope extends beyond LLMs to encompass generative AI at large by including diffusion-based image synthesis and multimodal architectures (where hallucination mechanisms differ fundamentally from those in autoregressive text generation). Second, rather than relying on a two-class typology, this review proposes a multidimensional taxonomy spanning seven classification axes (intrinsic vs. extrinsic, factual vs. logical, domain specificity, scope, intentional vs. unintended, detection difficulty, and temporality), supplemented by additional frameworks that address knowledge-repos-

itory interactions, self-contradictions, and prompt-induced errors. Third, the review integrates domain-level case studies from academic research, commercial deployment, and educational contexts that situate technical methods within their practical consequences—an aspect largely absent from prior surveys. Fourth, it consolidates the benchmark and metric landscape across six task categories into a unified mapping, and includes a glossary intended to lower the entry barrier for readers outside the core NLP community.

Building on the above observations and given the continued interest in GAI models, this review aims to examine hallucinations by systematically consolidating knowledge on their origins, detection, and mitigation strategies, and by tying diverse terminologies and frameworks to provide interested readers and researchers with a structured overview of state-of-the-art techniques. Moreover, this review examines detection methods that leverage specialized architectures and external knowledge sources, as well as correction mechanisms ranging from post-processing modules to integrated approaches (e.g., RAG and uncertainty calibration), and elucidates how model architecture, training data biases, and inference algorithms interact to produce factual errors. Beyond technical improvements, the review also addresses the ethical and social implications of hallucinations by emphasizing the need for systems that maintain user trust and proposing a comprehensive framework for benchmarking mitigation techniques using specialized datasets and metrics beyond traditional measures to facilitate reproducible experimentation and fair comparisons.

This review is structured as follows: Sect. 2 defines and explains the fundamental concepts and mechanisms behind hallucinations in GAI and establishes the necessary background for understanding this phenomenon. Building on this foundation, Sect. 3 introduces a structured classification of hallucination types. Section 4 then examines diverse strategies for detecting hallucinations in generated content, followed by Sect. 5, which dives into techniques for correcting and mitigating them. Section 6 outlines the benchmark tests and datasets commonly used to assess hallucinations in GAI and LLMs, to support the evaluation and comparison of such methods. The practical implications of hallucinations are highlighted in Sect. 7, which presents real-world examples from different domains and reviews corresponding mitigation efforts. Finally, Sect. 8 identifies promising avenues for future research and discusses the limitations of the current review to encourage continued exploration in this evolving field.

2 Understanding hallucinations in generative models

This section presents some of the key definitions and mechanisms behind hallucinations in GAI.

2.1 Defining hallucinations

Hallucinations in generative models are instances in which the output appears syntactically valid or visually coherent but lacks fidelity to factual or logical truth (see Fig. 1) (Chen et al., 2024d). These inaccuracies are not merely random errors but

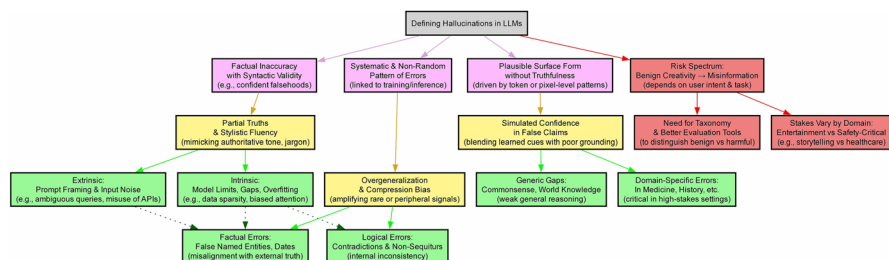


Fig. 1 Various definitions of GAI hallucinations

can represent systematic tendencies linked to the model's learning process, training data, or inference algorithms (Dziri et al., 2022a). While traditional models focused on reproducing statistical patterns without explicit checks for veracity, larger and more advanced systems were developed, and in such models, users/model developers began observing creative and unexpected fabrications that diverged sharply from real-world knowledge.

At a fundamental level, hallucinations differ from simple noise or random mistakes (Xu et al., 2025). They often exhibit a surface of plausibility that stems from leveraging token-level probabilities or pixel-level patterns that appear contextually consistent. This plausibility can be fueled by partial truths or subtle cues within the training data. For instance, a model might generate a text passage that mimics the style of authoritative articles that incorporate convincing references or domain-specific jargon, yet none of it accurately aligns with known facts (Kempt, 2025). In image generation, hallucinations may produce pictures with realistic compositions but embed artifacts or distortions that misrepresent reality (Sun et al., 2025a). The underlying mechanisms usually involve the model conflating different pieces of learned information or over-generalizing from limited examples. In such instances, cloaking factual errors under superficially coherent forms, similar episodes raise concerns about how well users can detect inaccuracies when the system exudes confidence.

2.2 Mechanisms behind hallucinations

LLMs typically rely on sequences of self-attention layers that capture sophisticated statistical dependencies across large data (Luo et al., 2023). These architectures, while adept at pattern recognition, do not inherently validate the factual content they produce. During training, the model is optimized to predict tokens based on context, but no direct objective penalizes it for producing coherent yet false statements. As a result, once training concludes, the model may generate plausible text snippets that deviate from established facts. This is often tied to the principle that hallucinations often arise from fundamental limitations embedded in model architectures and training protocols (see Fig. 2) (Xue et al., 2023). This shortfall becomes especially evident when the model faces prompts about niche topics or relies on partial context. In such instances, overfitting can amplify these tendencies: if the model memorizes idiosyncratic patterns or spurious associations from the training data, it can recombine them in ways that appear coherent but are factually unsound.

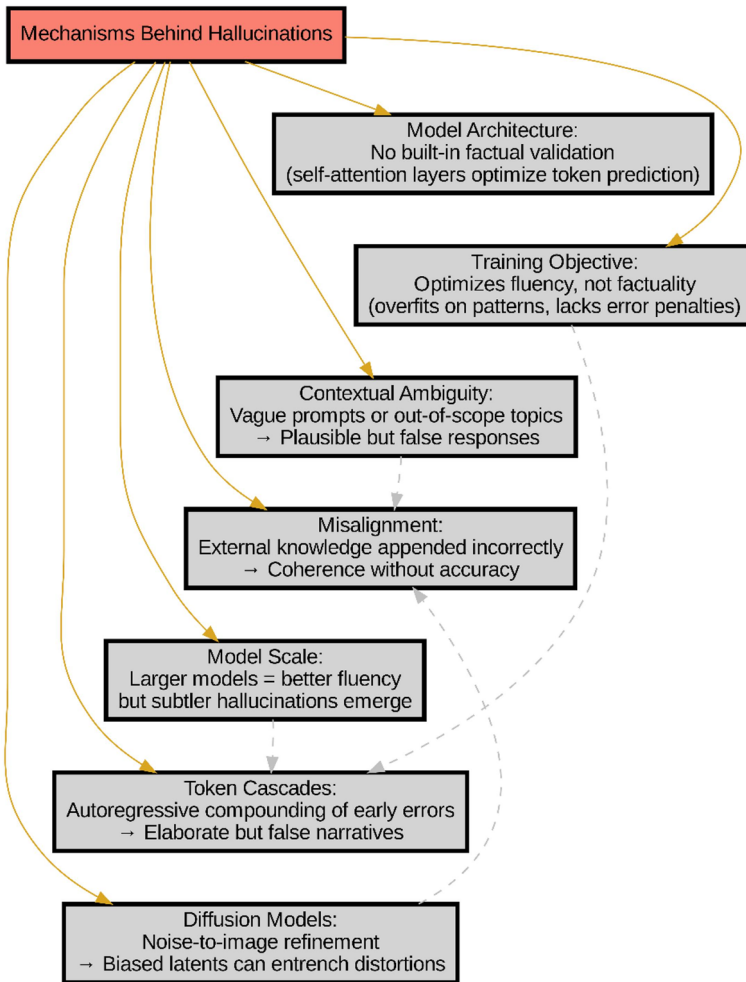


Fig. 2 Hallucination mechanisms

Contextual ambiguity also plays a central role in hallucinations (Perković et al., 2024). If a prompt is vague, contradictory, or implicitly demands knowledge beyond the model's training scope, the output may mix partial truths with fabrications to maintain the appearance of coherence (Rawte et al., 2024). Ambiguities can arise from user-provided texts or real-time data on which the model was never trained. In addition, misalignment between the retrieved data and the generative component can produce hallucinated outputs, which can indicate that simply appending external information does not resolve deeper representational gaps.

The model scale also interacts with hallucination tendencies. For instance, larger models often demonstrate greater fluency and broader domain coverage by reducing certain types of errors (while introducing subtler forms of misinformation) (Gonzales, 2025). Because these systems compress vast amounts of data into their parameter space, they become adept at pattern synthesis but remain susceptible to generating

statements that deviate from facts. Simply, the presence of extensive training data does not guarantee factual integrity if it contains biases, inaccuracies, or contradictions.

The token prediction process can compound errors when initial tokens deviate from factual statements (Jiang et al., 2025). Autoregressive models generate outputs token by token, each choice contingent on previous elements. A small factual slip early in the generation can trigger a self-reinforcing cascade, with subsequent tokens constructing an increasingly elaborate but incorrect narrative. The text's cohesiveness can mask the underlying errors, particularly for users who lack the domain expertise to recognize inconsistencies. When combined with high degrees of syntactic polish, these hallucinated narratives present a false confidence that can be misleading to users. Detecting and interrupting such cascades requires error-checking mechanisms that operate at both local (token-level) and global (document-level) scales, a non-trivial challenge given the complexities of natural language (Li et al., 2024).

Similar dynamics apply to diffusion models in the visual domain (Lan et al., 2024). Each iterative step in the diffusion process refines noise into an image, guided by learned patterns in shape, texture, and color (Lee et al., 2024). If the initial latent space representation or model parameters harbor biased or inaccurate assumptions, the refined image might exhibit subtle distortions that can lead to hallucinated details. Although state-of-the-art systems can produce strikingly photorealistic images, anomalies or factual misrepresentations arise when the model attempts to depict specific individuals, landmarks, or objects with verifiable properties. For instance, generating a historical figure's portrait may embed anachronistic attributes derived from incomplete or inconsistent training samples. The iterative nature of diffusion means that once an inaccurate detail appears in intermediate steps, the model can further entrench it in subsequent refinements.

As one can see, the mechanisms reviewed above continue to emphasize that hallucinations are not a simple glitch but rather an emergent property of how generative models learn, store, and retrieve complex data distributions (Lin et al., 2024).

3 Taxonomy of hallucinations

Establishing a taxonomy of hallucination-based errors is critical and warrants attention in both existing and future work in this area. The same is true of hallucinations, which span a wide spectrum (i.e., some errors are small and innocuous, while others dramatically mislead users or invalidate entire outputs). Therefore, this section presents a simple and systematic classification that allows interested readers to pinpoint the nature, cause, and impact of hallucinated content, which can inform more targeted mitigation approaches—see Fig. 3.

3.1 Core classification dimensions

3.1.1 Intrinsic vs. extrinsic hallucinations

One foundational axis of taxonomy hinges on whether a hallucination is *intrinsic* or *extrinsic* (Oelschlager, 2024). Intrinsic hallucinations arise from the model's internal

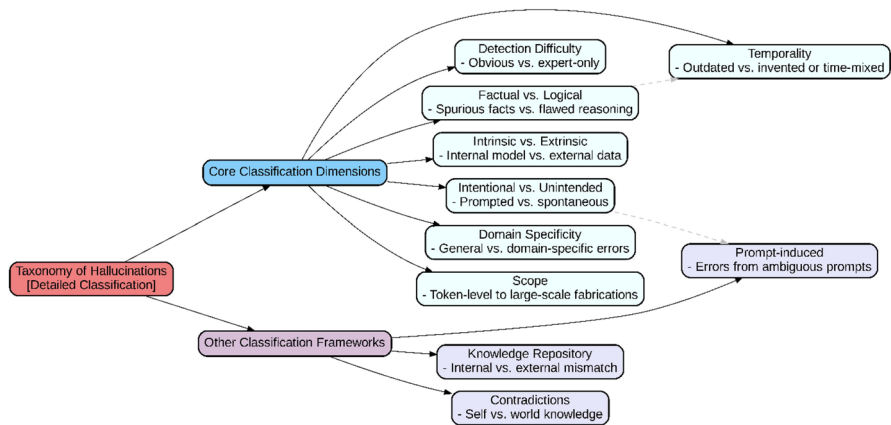


Fig. 3 Proposed taxonomy

parameter distributions and break when an unaligned or incomplete representation of the training data leads the system to fill knowledge gaps arbitrarily (Berbatova & Salambashev, 2023; Di et al., 2024). One might characterize intrinsic hallucination as a generative distribution that maximizes coherence with prior tokens, while failing to ensure that the output reflects any externally valid or cross-checked knowledge. Because these errors originate internally, they often elude simple detection if one relies solely on textual fluency or stylistic consistency (Snyder et al., 2024). In contrast, extrinsic hallucinations involve incorrect or contradictory usage of an external data source. Suppose a model consults a retrieval mechanism or knowledge graph but conflates references or picks segments from mismatched documents; the result can be an output that partially relies on actual data while embedding misleading segments (Lavrinnovics et al., 2025). This second category emphasizes verifying the chain of evidence linking the retrieved content to the final generated text. In practice, extrinsic hallucinations often occur in retrieval-augmented architectures that attempt to anchor responses in external knowledge yet inadvertently mismatch or misinterpret those sources.

3.1.2 Factual vs. logical hallucinations

A second dimension of classification distinguishes between *factual* and *logical* hallucinations (Huang, 2025). Factual hallucinations include spurious claims about named entities, dates, locations, or numerical data (Goyal & Durrett, 2020; Honovich et al., 2022). For instance, a language model may assert that a historical figure died in 1910, when in reality the individual passed away decades later. These errors are verifiable against authoritative references, which makes them comparatively straightforward to detect. Logical hallucinations, on the other hand, do not revolve around raw factual details but rather appear as contradictions within an argument’s reasoning structure. For example, a system might assert that a certain statement follows necessarily from a premise, even though careful logical analysis indicates no such derivation is valid. In text-based generative tasks, logical hallucinations appear in claims such as “if

A, then B must be true,” even though B has no logical connection to A. Such errors demand deeper analysis, as they may involve formal logic checks, proof-based tools, or inference frameworks that identify contradictions in multi-step reasoning. Such hallucinations can remain superficially coherent and often go undetected when users skim for explicit factual references alone.

3.1.3 Domain specificity of hallucinations

A related classification group is domain specificity. Broad-spectrum or general-purpose language models tend to produce *domain-agnostic* hallucinations, including random misstatements about widely known facts and contradictory statements about common concepts. These missteps typically arise from data coverage gaps or over-generalization in the model’s training data. By contrast, *domain-specific* hallucinations occur when a system is finetuned or optimized for a specialized field in which certain terms require meticulous accuracy. For example, an erroneous medical dosage or an incorrect citation to a legal precedent represents a domain-specific hallucination that can have pronounced negative consequences.

3.1.4 Scope of hallucinations

Another class can further categorize hallucinations by *scope*: from small-scale slips to large-scale fabrications. In this class, token-level confusion can manifest as the misplacement of an individual word, such as a date or name, which does not immediately disrupt the rest of the text but introduces subtle inaccuracies. At the other extreme, entire passages/arguments may be constructed on a spurious foundation and can reveal a large-scale hallucination that permeates the entire output. Intermediate cases involve incorrectly attributing quotes or references, an error that can undermine the legitimacy of a single paragraph while leaving the surrounding text technically sound. This scope-based classification is especially relevant to designing detection and correction algorithms, as methods that excel at locating small token-level inconsistencies (e.g., referencing 1893 instead of 1983) may be ill-suited to handling large blocks of text that rely on fundamentally flawed premises.

3.1.5 Intentional vs. unintended hallucinations

Intentional versus *unintended* hallucinations represent a critical boundary in model behavior that significantly impacts detection and mitigation strategies (Grover et al., 2024). Intentional hallucinations emerge when models are explicitly or implicitly prompted to generate content beyond their knowledge constraints (i.e., when users request information about fictional entities or hypothetical scenarios or when the phrasing of prompts [e.g., “imagine that,” or “create a story about”]) signals permission to depart from factual constraints. In these scenarios, the model engages in what might be termed “sanctioned hallucinations,” in which generating non-factual content aligns with user expectations. This contrasts with unintended or spontaneous hallucinations, which occur when a model produces non-factual information without prompting while operating under the implicit expectation of factual accuracy. These

spontaneous hallucinations frequently arise from internal activation patterns or distributional biases in the training data that cause the model to confidently present plausible-sounding but unfounded claims. The distinction carries significant implications for evaluation frameworks, as the same text generation might be considered appropriate in contexts where creative elaboration is expected, yet deeply problematic when factual reporting is the implied contract between system and user (Reddy et al., 2024). This dimension also intersects with user experience considerations, as users may develop mistaken mental models of a system's capabilities when intentional and unintentional hallucinations are not clearly differentiated in the interaction design.

3.1.6 Detection difficulty of hallucinations

The *detection difficulty* category of hallucinations spans a spectrum from immediately apparent errors to nearly undetectable falsehoods that resist verification without specialized knowledge or extensive cross-referencing (Kim et al., 2024). At the most easily detectable end lie what might be termed *obvious* hallucinations (i.e., statements containing internal contradictions, claims that violate common knowledge, etc.). These errors can often be caught through basic consistency checks or by non-expert users with minimal domain knowledge. The middle range encompasses *plausible* hallucinations that present no obvious markers of incorrectness and maintain internal coherence. However, these contain factual errors that require specialized knowledge to identify (e.g., a misattributed scientific discovery that would seem plausible to anyone outside that discipline). At the most challenging end of the spectrum lie *expert-level* hallucinations that can only be identified by individuals with domain expertise, specialized tools, or access to authoritative references (e.g., mathematical proofs with non-obvious errors, etc.). This class of hallucinations has severe implications for model deployment safety, as the proportion of difficult-to-detect hallucinations may remain hidden during standard quality assessments yet appear as significant liabilities in real-world applications (where users lack the expertise and incentive to critically evaluate model outputs).

3.1.7 Temporality of hallucinations

The *temporal* dimension of hallucinations distinguishes between *anachronistic* representations of previously valid information versus wholly *fabricated* content with no historical validity. Temporally outdated hallucinations occur when models present information that was once accurate but has since been superseded by developments, updates, or changes (e.g., citing an outdated government official or describing a legislative framework that has been amended). These hallucinations stem from training data cutoffs, where models lack access to recent developments, or from biases toward older, more established information that may have greater representation in the training data. In contrast, fabricated hallucinations involve content that was never accurate at any point in time (e.g., falsely attributed quotes, etc.). A more nuanced subcategory includes temporal conflation hallucinations, in which elements from different time periods are erroneously combined (for instance, describing historical figures as contemporaries when they lived centuries apart). The temporal dimension

interacts significantly with types of factual hallucinations but introduces unique verification challenges, as temporal accuracy requires checking not only whether information exists but also whether it is valid or makes sense within the specific timeframe referenced or implied in the generated content (Zhou et al., 2024b). This dimension holds particular relevance for applications in rapidly evolving domains, where the half-life of factual accuracy may be measured in days or weeks rather than years.

3.2 Other classification frameworks

While the above covers the most commonly used classifications, we now shed more light on some less commonly known categories. It is worth noting that, in some instances, these categories may sound like unique/special cases arising from commonly known classifications.

3.2.1 Knowledge repository interactions

This class of hallucinations focuses on how the hallucination interacts with external knowledge repositories (Huang, 2025). In modern model architectures, a generative model integrates memory modules or retrieval operations to ground its outputs in real-time data (Leiser et al., 2023). This approach, known as retrieval augmentation, can reduce hallucinations but also introduces new avenues for error: if the retrieval module is incomplete, stale, or incorrectly ranked, the model's subsequent text might reflect the inadequacies of the retrieved data. A well-defined taxonomy accounts for whether a hallucination stems from an internal mismatch (the system never consulted the knowledge base) or an external mismatch (the knowledge base was consulted but incorrectly interpreted). This distinction significantly impacts the type of remedial strategies likely to be successful.

3.2.2 Contradictions in generated content hallucinations

Some may also classify hallucinations by the presence of contradictions within a single output (as opposed to across multiple outputs). For instance, an internally self-contradictory statement might assert "the capital of Canada is Toronto" in one sentence, followed by "the capital of Canada is Ottawa" in another. Such *self-conflicts reflect* an absence of coherent consistency checks, which can be traced to the model's token-by-token generation. Yet, the same error might remain hidden to a casual reader if the contradictory sentences are sufficiently far apart in the text. Further, another type of contradiction arises when the generative system outputs a statement that directly counters widely accepted world knowledge but maintains local coherence. This might be labeled a *world-knowledge contradiction*, distinct from self-conflicts that pit different model outputs against one another. A clear taxonomy acknowledges that these two subtypes require different detection approaches: self-consistency checks or local N -gram scanning for self-conflicts versus reference-based or retrieval-based checks for world-knowledge contradictions.

3.2.3 Prompt-induced hallucinations

Another dimension pertains to how the user *prompt* or context influences the emergence of hallucinations (Rawte et al., 2023b). If a user query is ambiguous or contradictory, the model may guess the user's intent and thereby embed fictitious details to fill the gap. In such scenarios, the hallucination is partially user-driven, as it arises from the system's attempt to maintain a veneer of coherence despite incomplete or conflicting instructions. Differentiating these prompt-based hallucinations from those arising in well-formed queries clarifies whether developers should focus on refining prompt handling or on reinforcing the model's internal validation. In that sense, the taxonomy contemplates whether the motivation for incorrect content lies in the provided textual environment or the system's training procedure. This distinction is valuable in interactive applications, where user clarifications or disambiguation requests can significantly reduce spurious content if the system is designed to ask clarifying questions rather than forging ahead with minimal context.

3.2.4 Self-awareness and disclaimers

A final dimension to consider involves the presence or absence of the model's *self-awareness or disclaimers* during generation (Azaria & Mitchell, 2023). In some settings, generative systems are taught to disclaim uncertainty or cite data sources when relevant. If such disclaimers are consistently absent or incomplete, the model's illusions become more hazardous, as its output can appear confidently authoritative even when factually void. On the other hand, a system that regularly flags questionable statements or offers disclaimers about incomplete knowledge may produce less harmful illusions because users are repeatedly forewarned. The above taxonomy can thus incorporate whether the model is configured to express uncertainty or disclaimers when venturing beyond validated territory. This notion is not solely about how the system is perceived; it also influences detection and correction strategies, as disclaimers often signal potential pivot points where more thorough fact-checking is warranted or beneficial.

4 Detection strategies for hallucinations

The search for methodologies for hallucination detection has attracted significant attention in generative systems (Chen et al., 2023b; Heo et al., 2025; Huang, 2025). This section reviews post hoc detection procedures, uncertainty estimation frameworks, retrieval-based checks, and comparative evaluations that consolidate various methodologies. Although these methods vary substantially in their theoretical foundations and computational footprints, their collective motivation rests on improving the reliability of generative models by distinguishing plausible fabrications from genuine, evidence-based responses (Bai et al., 2024).

It is also worth noting that a consistent theme in detection research is the diversity of error categories, wherein each demands specialized methods for identification (Chakraborty, 2025). Thus, some detection frameworks target gross factual errors,

while others examine subtler phenomena like inconsistent style or covert contradictions spanning multiple sentences. This multiplicity highlights the complexity of designing universally applicable techniques.

4.1 Post-hoc detection techniques

An extensive category of detection methods focuses on analyzing generative outputs after they have been produced (Underwood & Fenwick, 2024). These post-hoc approaches generally do not alter the generation process itself; rather, they rely on separate modules or workflows to scrutinize textual, visual, or multimodal content (Yu et al., 2024). One technique centers on consistency and coherence checks that measure whether the output's internal logic remains stable when examined sentence by sentence or region by region in an image. This is based on the notion that models often emit contradictory statements when hallucinating, so detection methods that track semantic or conceptual alignment can flag inconsistencies (Liu et al., 2025a). Such checks can be performed by training an auxiliary classifier that takes a piece of generated text as input and predicts whether it is prone to factual errors or self-contradiction. These classifiers leverage large-scale curated datasets of correct versus incorrect sequences to discern patterns associated with erroneous content.

Another widely discussed post-hoc strategy involves comparing generated text against external reference datasets and fact-checking tools (Ravichander et al., n.d.). GAI systems can be configured to extract named entities, citations, or numerical claims from the model output, then query an external knowledge base to verify that these elements align with established facts. Thus, if a generated statement conflicts with high-confidence entries in a reputable database, the system flags it for further review. Such fact-checking frameworks often integrate specialized retrieval routines and natural language processing architectures that parse the text into discrete claims (Nahar et al., 2025). Each claim is then validated by cross-referencing domain-specific data, news archives, or structured data repositories. Such mechanisms have become especially common in digital journalism, where rapid verification of sources remains a priority.

Contrastive learning has also gained traction as a method for detecting hallucinations post-generation (Jiang et al., 2024). In these setups, a model learns to distinguish between *true* and *fabricated* versions of a text or image, typically through a specialized encoder that outputs embeddings (Fei et al., 2024). The encoder learns representation boundaries indicative of deceit or invention by creating paired examples in which one version is deemed accurate, and the other is artificially manipulated. At inference time, the encoder processes the newly generated output and assigns it a score reflecting how closely it aligns with the distribution of genuine examples. High-dimensional embeddings thus become a powerful signal for evaluating whether the content is realistically grounded or straying into hallucinatory patterns (Chuang et al., 2024).

In addition, natural language inference (NLI)-based detection mechanisms are related to contrastive learning (Lei et al., 2023). NLI models have a history of being trained to determine if a hypothesis is entailed, contradicted, or neutral with respect to a premise. From this review's lens, model developers have repurposed these sys-

tems by treating the generative model's output as the hypothesis and an authoritative statement (or curated reference) as the premise. If the NLI model predicts a contradiction, the system infers that the text likely contains a hallucinatory element. This process assumes that a reliable ground-truth statement is readily available or can be dynamically assembled, which is not always the case. Nonetheless, NLI strategies can serve as an effective post-hoc filter for domains with canonical definitions and verified reference sentences.

Specialized verification models trained specifically to identify hallucinations represent another post-hoc approach (Oh et al., 2024). Here, and instead of relying solely on fact-checking tools or general-purpose classifiers, verifiers are optimized on datasets containing known hallucinations. Such training allows them to capture domain- or model-specific patterns of error. For instance, a tailored verifier could learn to detect those merges if a language model is prone to mixing multiple historical events when generating timeline narratives. This targeted optimization can outperform broader detection methods that may not be attuned to the nuances of a particular domain. However, building specialized verifiers requires substantial labeled data, and the approach's reliability is directly tied to how comprehensively the training data represents the spectrum of possible hallucination instances.

4.2 Uncertainty calibration methods

A complementary direction in detection research focuses not on analyzing final outputs but on how models internally estimate their likelihood of correctness as they generate tokens or images (Simhi et al., 2025). Thus, uncertainty calibration aims to align a model's expressed confidence with its actual reliability, thereby enabling the system to signal when it is unsure (Chrysos et al., n.d.). Such techniques for quantifying model confidence generally rely on output distributions, gradient magnitudes, or other internal metrics that approximate the model's certainty about a particular token or pixel sequence. In such a technique, properly calibrated models can raise alerts when they detect unusually high entropy or conflicting signals as a means to suggest that the produced content may contain fabricated elements.

One such example can be seen in the case of Bayesian approaches (Wang et al., 2023b). These approaches treat the model's parameters or outputs as random variables with associated probability distributions. The system arrives at a more robust estimate of confidence by sampling different plausible parameter sets or employing ensemble methods that average predictions from multiple networks. If the ensemble members diverge significantly, the generation is flagged as suspect, indicating that the model has not converged to a consistent solution. This strategy is particularly relevant for domains where ground-truth data is sparse or where a single model instance may struggle to capture all nuances. Bayesian uncertainty estimation, however, can be computationally intensive, as repeated inference or parameter sampling can significantly increase the time and memory required. As a result, practitioners often consider it for specialized tasks where reliability supersedes computational cost.

Self-consistency checks and multiple sampling are lighter approaches to uncertainty estimation (Cheng et al., 2024). Instead of drawing from Bayesian principles, the model simply generates multiple candidate outputs for the same input. The results

are then compared to identify whether a consensus emerges. If a high degree of variability exists among these candidates, it implies that the model's internal representation remains uncertain, which, by extension, makes it more prone to hallucinations (Yuan et al., 2024). Temperature scaling also modulates the deterministic nature of the generation process (Uluoglakci et al., 2024). In this approach, a high temperature fosters more randomness, whereas a low temperature enforces conservative, high-probability token choices. Intermediate temperature settings can offer a trade-off, and it is worth stressing that even extremely conservative outputs can still contain factual slips if the underlying knowledge is incomplete.

Token-level uncertainty estimation refines the above concept by evaluating the confidence attached to each token during generation (Krishnan et al., 2024). Some methods compute the differential change in model logits as subsequent tokens are selected to identify points at which the model appears indecisive (Ma et al., 2025). If confidence dips sharply for crucial tokens (as indicated by the numeric data), this may indicate that the surrounding text is speculative. Implementations in computer vision rely on analogous metrics that focus on pixel-level or latent-space confidence. Thus, when highlighting specific regions in a generated image where the model's confidence is low, a model developer can localize potential fabrications. Such token- or pixel-level insights are useful for interactive or semi-automated verification workflows, as they direct attention to the portions of the output most in need of additional checks.

Another benefit of uncertainty calibration is the opportunity for early intervention (Savage et al., 2025). If the system can detect moments of confusion during the generation process, it can either solicit external data, revise the prompt, or initiate self-correction modules. This dynamic approach is distinct from post-hoc pipelines that examine outputs only after generation is complete. In a way, model developers can integrate confidence signals directly into the generative mechanism by building architectures that incorporate fallback strategies (e.g., narrower retrieval queries or default disclaimers) when uncertainty surpasses a threshold. Although uncertainty calibration does not fully eliminate hallucinations, it provides a nuanced gauge of when the model is more likely to produce ungrounded information.

4.3 Retrieval-augmented generation (RAG)

While post-hoc and uncertainty-based strategies focus on identifying hallucinations either after the fact or through internal signals, RAG takes a more proactive approach by guiding the model's reasoning with external data (Bacciu et al., 2025; Song et al., 2024). These methods integrate curated knowledge bases at runtime to allow the generation module to consult up-to-date documents, structured entries, or other verifiable content when forming responses (Fan et al., 2024). The fundamental premise is that a generative architecture that continually references external evidence is less likely to fabricate facts. By design, it can cross-check internal representations against stable information sources before finalizing each token or pixel segment.

The operational flow in RAG typically involves a dual-step process: first, a retrieval engine or knowledge retriever identifies relevant documents or snippets based on the user's query or partial generative context (Bécharde & Ayala, 2024). Then, the gen-

erative component conditions on these retrieved elements, either through attention mechanisms or specialized fusion modules, and combines the retrieved representations with the model's internal hidden states. Some architectures, such as Fusion-in-Decoder, extend these ideas by performing multi-step reasoning and iteratively refining the retrieved context as the generation proceeds (Reddy et al., 2024). In parallel, techniques such as FLARE incorporate signals from both sparse (keyword-based) and dense (neural embedding-based) retrieval systems to maximize coverage of relevant data (Arakelyan et al., 2024; Su et al., 2024). The synergy between these retrieval vectors can refine the generative process, as each offers distinct advantages in capturing domain-specific or lexical nuances.

Knowledge graph integration resembles another structured approach for fact verification within the RAG technique (Barry et al., 2025; Procko & Ochoa, 2024). This integration links generated tokens or image features to nodes and edges in a graph such that the system can track the logical connections among factual information. If the generative model attempts to produce an entity or relationship not supported by the graph, it can be flagged or corrected. The challenge, however, lies in constructing and maintaining high-quality graphs that accurately mirror reality, as minor inconsistencies in the graph can lead to erroneous flags or missed detections. Furthermore, balancing the retrieval of graph substructures with the generative model's computational constraints requires careful engineering to ensure latency remains manageable for real-time deployments (Ribeiro et al., 2022).

4.4 Comparative analysis of detection methods

Having surveyed an array of detection approaches, it can be of merit to compare them on qualitative and practical grounds (as each method exhibits unique strengths as well as limitations that might make it more or less suitable for particular scenarios)—see Table 1.

For example, post-hoc techniques excel at modular integration and hence allow organizations to overlay them on existing generative infrastructures without retraining the core model. They also accommodate independent updates to external fact-checking or verification components. However, these methods typically rely on thorough scanning of outputs, which can be computationally expensive and may delay real-time responses. On the other hand, uncertainty calibration offers a more organic approach to detection by flagging potential inaccuracies directly during generation. This real-time indication of confusion can facilitate immediate intervention or fallback mechanisms, potentially minimizing the spread of erroneous outputs. Yet, one must carefully calibrate these signals, as overly pessimistic settings might trigger false alarms and hamper system usability. Ensemble and Bayesian strategies further boost robustness but require considerable computational resources, which raises natural concerns over scalability.

As seen above, RAG, by contrast, aims to minimize hallucinations by providing relevant external knowledge. When executed properly, this tactic can drastically enhance factual precision (which makes RAG attractive for domains where accuracy is essential). Nonetheless, integrating retrieval systems introduces complexities, including relevance filtering, version control for knowledge repositories, and

synchronization between retrieval and generation modules. Therefore, improper alignment or partial retrieval can still yield speculative outputs. Moreover, retrieval augmentation presupposes that high-quality references exist, which may not be the case for emerging or niche topics. Moreover, the GAI system must reconcile contradictory or outdated information within the retrieved set. Strategies such as iterative read–verify loops or rank–select pipelines have emerged to mitigate these issues, though each step introduces additional complexity (Zhao et al., 2024). Another concern arises in domains with no authoritative external database or heavily fragmented data. In these scenarios, retrieval augmentation risks offering incomplete or scattered references, limiting its capacity to eradicate hallucinations.

From a practical standpoint, computational overhead versus accuracy is a core trade-off. Post-hoc verification that relies on large external data can significantly raise latency, especially if fact checks or inferences from specialized classifiers are performed for every segment of generated text. Uncertainty calibration can be executed on the fly but demands additional sampling, parameter expansions, or model ensembles. Meanwhile, RAG strategies require frequent lookups and sophisticated ranking heuristics, which can become bottlenecks when the underlying databases are large or usage volumes are high. Each method’s feasibility, therefore, hinges on factors such as available hardware, the volume of user queries, and tolerance for potential slowdowns.

5 Correction and mitigation strategies

Mitigating hallucinations involves more than merely labeling or identifying suspicious output (Roy & Roy, 2025). Instead, remedial techniques also strive to intervene in the generation loop or refine the underlying parameters to enhance factual adherence (Yu et al., 2024; Yadkori et al., 2024). Such interventions can occur before the model produces a response, during token generation, or after the completion of output, sometimes looping back to revise text or imagery once errors are flagged. This section sheds light on a variety of correction and mitigation mechanisms, as well as the pragmatic trade-offs that must be weighed when implementing these tools in real-world pipelines (see also Fig. 4 and Table 2).

5.1 Proactive correction techniques

Early interventions during training or fine-tuning can substantially reduce the model’s propensity to generate spurious content. One approach involves retraining the model on specialized datasets with validated facts and vetted references (Sood et al., 2025). Under this approach, rather than allowing the model to consume large volumes of training data indiscriminately, developers curate and filter their data repositories so that verified factual sources—such as academic publications and peer-reviewed materials—form the primary basis for the training process. The assumption here is that a model steeped in reliable information is less inclined to invent details. Although this can mitigate hallucinations, it also restricts the breadth of the model’s potential knowledge if the dataset is overly narrow.

Table 1 Summary and comparison of the reviewed detection strategies

Detection strategy	Approach	Key techniques	Strengths	Weaknesses	Examples	Computational requirements	Domain applicability	Comparative advantages
Post-hoc detection techniques	Analyzing outputs after generation without altering the generation process	- Consistency and coherence checks - External fact-checking - Contrastive learning - Natural Language Inference (NLI) - Specialized verification models	- Modular integration with existing systems - No need to re-train core models - Can accommodate independent updates to verification components - Works with any generative model	- Computationally expensive - May delay real-time responses - Requires thorough scanning of outputs - Limited to detecting errors that match known patterns	- Auxiliary classifiers trained on correct vs. incorrect sequences - Entity extraction + knowledge base queries - Embedding-based comparison to genuine examples - NLI models treating generated output as hypothesis against ground truth premise	Medium to High (Depends on verification complexity)	- Broad applicability in domains with clearly defined facts - Suitable for text, images, and multimodal content	- Best for systems where modification of the generation process isn't feasible - Excellent for platforms requiring transparency in verification - Superior for batch processing over real-time applications
Uncertainty calibration methods	Estimating model confidence during generation to flag potential errors	- Bayesian approaches - Self-consistency checking - Multiple sampling - Temperature scaling - Token-level uncertainty estimation	- Real-time indication of model confusion - Facilitates immediate intervention - Can minimize spread of errors - Provides nuanced gauge of reliability - Enables early intervention	- Requires careful calibration to avoid false alarms - May hamper system usability if too conservative - High computational cost for ensemble methods - Scalability concerns	- Ensemble methods averaging predictions from multiple networks - Generating multiple outputs and checking for consensus - Analyzing differential change in model logits - Pixel-level confidence metrics in image generation	High (Especially for Bayesian and ensemble approaches)	- Versatile across domains - Particularly valuable in high-risk applications - Effective for both structured and open-ended tasks	- Superior to post-hoc methods for real-time applications - Better at signaling uncertainty in novel situations compared to RAG - More integrated with generation process than post-hoc methods

Table 1 (continued)

Detection strategy	Approach	Key techniques	Strengths	Weaknesses	Examples	Computational requirements	Domain applicability	Comparative advantages
Retrieval-augmented generation (RAG)	Proactively guiding generation with external data to prevent hallucinations	- Knowledge retriever+ generative component - Fusion-in-Decoder - FLARE (combining sparse and dense retrieval) - Knowledge graph integration - Iterative read-verify loops	- Significantly enhances factual precision - Reduces burden on model parameters - Provides transparency through referenced documents - Anchors outputs in empirical reality - Knowledge-centric approach	- Complex integration requirements - Relies on high-quality external references - Challenges with contradictory/outdated information - Relevance filtering difficulties - Less effective for emerging/niche topics	- Conditioning generation on retrieved documents - Knowledge graph verification during generation - Dual-step retrieval and generation process - Multi-step reasoning with iterative refinement - Rank-select pipelines for information reconciliation	Medium to High (Depends on knowledge base size and retrieval complexity)	- Excellent for well-documented fields - Less effective in areas with limited authoritative data - Strongest in domains requiring strict factual accuracy	- Most proactive approach to preventing hallucinations - Better factual grounding than uncertainty methods - More efficient than post-hoc for established knowledge domains

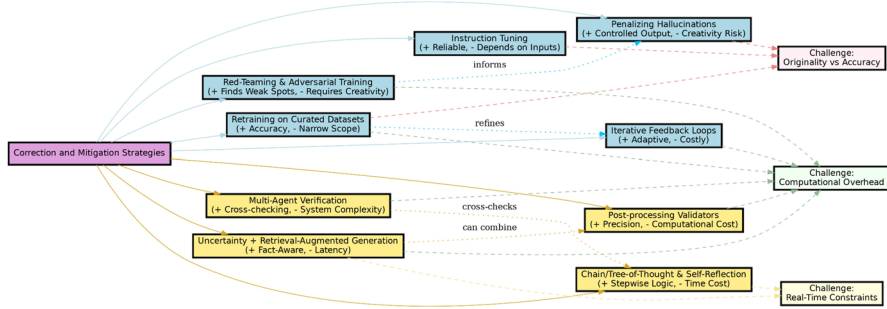


Fig. 4 Illustration of correction and mitigation strategies

Model refinement through iterative feedback loops offers another pathway to proactive correction (Cai et al., 2025). In such systems, human evaluators or specialized software agents inspect intermediate or final outputs and provide corrections or critiques. The model then incorporates these signals into subsequent training rounds to gradually align its responses with factual references. This iterative process may be facilitated by reinforcement learning that adapts parameters based on reward signals tied to veracity (Pternea et al., 2024). This repeated exposure to corrected content forces the model to internalize certain knowledge structures more robustly. One should note that the approach can be resource-intensive, as continuous feedback from domain experts might be necessary for complex topics (Gosmar & Dahl, 2025). Nevertheless, this process yields a dynamic correction system that evolves with the model's usage patterns and newly uncovered knowledge gaps.

An additional technique involves adjusting training objectives to penalize hallucinations (Tang et al., n.d.). One example is seen in constitutional AI frameworks, which aim to embed constraints or rules into the learning algorithm to keep the model's outputs within the bounds of factual accuracy or ethical guidelines (Cheng, 2025). These guidelines can be encoded into a specialized objective function or integrated into a multi-objective optimization scheme to effectively reshape the loss landscape and nudge the model away from ambiguous or purely pattern-based responses. While these frameworks show promise for constraining extreme outputs, they must be carefully calibrated to avoid stifling the model's overall expressiveness. Furthermore, the diversity of factual norms across different cultural or professional contexts complicates the universal application of rigid constraints.

Another robust approach to reducing hallucination is red-teaming and adversarial training (Liu et al., 2025b). Red-team exercises involve intentionally challenging a model with prompts designed to elicit imaginative or deceptive responses. This way, model developers identify vulnerabilities or recurring hallucination triggers by systematically exposing the system to these adversarial scenarios. These insights then feed back into specialized training loops, in which the model is penalized for producing erroneous outputs when prompted adversarially (Gokcimen & Das, 2025). Over successive iterations, the model becomes more adept at resisting prompts that lure it into hallucinations. A salient advantage of red-teaming is its ability to uncover corner cases that might remain obscure during conventional testing. However, this method

Table 2 Comparison of hallucination correction and mitigation strategies in GAI

Correction strategy	Core approach	Key techniques	Strengths	Weaknesses	Examples	Computational requirements	Domain applicability	Comparative advantages
Proactive correction techniques	Intervening during training or fine-tuning to reduce hallucination propensity	- Specialized dataset retraining - Iterative feedback loops - Adjusted training objectives - Red-teaming and adversarial training - Instruction tuning	- Addresses hallucinations at the source - Creates models with inherently better factual grounding - Can significantly improve overall reliability - Prevents hallucinations before they occur - More robust long-term solution	- Resource-intensive training processes - Requires carefully curated data - May restrict model's knowledge breadth - Balancing accuracy vs. generative creativity - Training-time interventions can not adapt to new topics	- Training on academic sources and peer-reviewed materials - Human evaluator feedback incorporated into training - Constitutional AI frameworks with factual constraints - Systematically challenging models with deceptive prompts - Training with instructions requiring factual grounding	High (Especially for complete retraining)	- Broad applicability across domains - Most effective when training data covers the domain thoroughly - Works best for stable knowledge domains	- Superior long-term solution compared to reactive approaches - More fundamental correction than post-generation fixes - Creates models that require less runtime intervention
Hybrid approaches	Combining detection and correction elements to create multi-layered solutions	- Uncertainty calibration + RAG - External validators with feedback loops - Multi-agent verification systems - Chain-of-thought prompting - Tree-of-thought prompting - Self-reflection techniques	- Flexible implementation options - Can address different error types simultaneously - Creates robust feedback cycles - Allows for iterative improvement - Combines complementary strengths of different approaches	- Complex integration requirements - Potentially high computational cost - Multiple passes may be required - May introduce significant latency - Orchestration challenges between components	- Systems that retrieve external data when confidence is low - Cyclical feedback processes with automated correction modules - Multiple models cross-examining each other's outputs - Breaking reasoning into validated incremental steps - Model-driven critique of its own outputs	Medium to Very High (Depends on complexity of integration)	- Versatile across domains - Particularly effective for complex reasoning tasks - Valuable for critical applications requiring multiple verification layers	- More flexible than purely proactive approaches - More thorough than single-strategy methods - Better adaptation to novel scenarios than fixed training - Can combine strengths of multiple approaches

hinges on the creativity/thoroughness of the red-teamers since unmapped prompts could still bypass the model's defenses.

Instruction tuning for hallucination reduction refines the system's interpretation of user directives (Si et al., 2025). Here, instructions are tuned by their developers to require factual grounding through referencing verifiable data or inserting disclaimers when the answer remains uncertain. Such conditioning internalizes the model behaviors to correctness. Although instruction tuning can significantly improve reliability, it presupposes that the instructions themselves are well-formulated and consistently enforced (Niess et al., 2025). For example, if a user supplies contradictory instructions or real-time usage deviates from training conditions, the model may revert to guesswork. Nonetheless, combining such tailored instructions with the detection strategies discussed earlier can establish robust feedback loops that minimize hallucinations.

5.2 Hybrid approaches

While proactive measures can reduce hallucination rates, some scenarios require multi-layered solutions that integrate detection and correction (Ji et al., 2023). One example is merging uncertainty calibration with RAG to create a system that consults external sources when its internal confidence drops. Under this hybrid arrangement, the model first evaluates whether its token predictions are robust. If it senses significant confusion, it triggers a retrieval query to gather supplementary facts before continuing. The retrieved data can then rectify misleading tokens mid-stream to bridge the gap between purely parametric and knowledge-centric modeling (Galitsky, 2025).

In other cases, external validators act as post-processing filters that examine the generated text for suspect claims (Shi et al., 2025a). Some implementations rely on large external data, while others incorporate domain-specific rule sets or neural modules trained to correct particular classes of errors. For example, if a suspicious segment is detected, an automated correction module either modifies the output directly or instructs the generative model to re-generate specific parts (Leiser et al., 2024). This method operates as a cyclical feedback loop until the validator's checks meet a certain threshold. Therefore, its computational cost can be high, as multiple passes may be required, but the end result is a more refined output that aligns with validated facts.

A multi-agent verification system extends this idea by orchestrating multiple generative or discriminative models to cross-examine each other's outputs (Sun et al., 2025b). In essence, one model might produce a candidate solution, while another model, potentially specialized in fact-checking, critiques/evaluates the output. If the second model identifies inaccuracies, it proposes corrections that the first model can integrate (Shi et al., 2025b). This tactic can be particularly beneficial when the models involved have complementary strengths, such as a LLM with broad coverage and a smaller but more precise domain-focused verifier.

Chain-of-thought prompting and tree-of-thought prompting exemplify other modern/advanced techniques that encourage the model to break down reasoning processes into incremental steps (Hu et al., 2025; Zhao et al., 2023). Thus, instead of generating a final answer immediately, the system delineates intermediate justifications or sub-questions that must be resolved en route to a conclusion. These intermediate steps

can be validated separately, either by the model itself or by an external checker. In this case, if a particular step is not factually plausible, the entire chain is flagged for re-evaluation. This iterative, multi-stage reasoning framework mitigates the risk of unbounded creative leaps that lead to hallucinations. In the same vein, self-reflection techniques prompt the model to critique or explain its own outputs by requesting an internal analysis of the reasoning chain. This way, it becomes possible to detect logical or factual vulnerabilities early to reduce the chances that a flawed line of reasoning goes uncorrected.

5.3 Challenges in implementing correction strategies

Despite the innovative array of the above mitigation techniques, practical barriers remain (Roustan & Bastardot, 2025). One fundamental tension involves reconciling the need for originality with the imperative for factual accuracy. For example, creative writing applications or ideation tools often leverage the generative model's capacity to synthesize novel ideas. On the one hand, overzealous constraints might diminish spontaneity and lead to a model that favors bland or heavily restricted outputs (He et al., 2025). In a way, and as one might expect, determining the precise extent of correction depends on context. This tension becomes especially pronounced in user-facing applications that cater to heterogeneous user needs, which may range from mere exploration to mission-critical analysis.

Computational overhead is another factor that can affect the selection of correction strategies (Belyi et al., 2025). This is due to the fact that many of the above correction methods, especially those involving repeated queries to knowledge bases or iterative generation loops, can impose significant latency (Zellinger et al., 2025). Therefore, such methods become problematic during deployment (where response speed shapes user satisfaction). Techniques that involve large ensembles or repeated sampling place further strain on the computing infrastructure, raising questions about how to scale to millions of daily requests. Even offline or batch processes can become unwieldy if the dataset is large and each sample requires multiple rounds of verification or retrieval. Model developers must therefore weigh the performance penalty against the value of higher reliability and tailor solutions to the available computational resources.

Similarly, real-time performance constraints further complicate matters, as seen in rapid interactions essential to dynamic interfaces (e.g., voice assistants). In such interfaces, if the system's correction framework is too time-consuming, users may abandon it in frustration (Huq et al., 2025). Therefore, prioritizing a subset of corrections or deploying a two-phase pipeline (i.e., a system in which more thorough checks follow quick responses if time permits) may represent a possible compromise. Another approach involves partial caching or indexing of frequently accessed knowledge to speed retrieval cycles (Gao et al., 2025). In most cases, there is a need to achieve a delicate equilibrium between prompt responsiveness and robust verification, which remains an active area of research, particularly for heavily trafficked public interfaces.

6 Benchmark tests and datasets for hallucination evaluation

A diverse set of benchmark tests and datasets has been developed over the years to systematically evaluate hallucinations in GAI and LLMs (Bao et al., 2024; Hu et al., 2024b; Liu et al., 2024a). These resources include publicly available academic datasets and proprietary internal test suites that address a range of tasks, from factual question answering to multimodal content generation. From this lens, this section provides an overview of major datasets commonly used to detect, characterize, and mitigate hallucinations. We provide an overview of how these datasets are structured, annotated, and intended for use, along with an analysis of their known limitations. In addition, this section discusses the most commonly used metrics for quantifying how consistently models adhere to factual information—see Fig. 5 and Tables 3 and 4.

6.1 Factual question answering benchmarks (closed-book QA)

Closed-book or open-domain question answering (QA) benchmarks are a primary means of assessing an LLM's ability to correctly recall factual information without recourse to external context at inference time (Huo et al., 2023; Zhuang et al., 2023). A prominent family of benchmarks includes Natural Questions (NQ) (Lee et al., 2019) and TriviaQA (Joshi et al., 2017), which offer large collections of questions that reflect real user queries or trivia-style prompts. Each question is accompanied by short factual answers, often focusing on entity attributes such as names and dates. Scoring is typically based on exact-match accuracy, and any substantial deviation from a verified ground truth is classified as an incorrect or hallucinated output. Poor performance on these benchmarks, especially when incorrect details are presented as facts, suggests that the model may hallucinate. Although these benchmarks offer valuable insights into the parametric knowledge embedded in an LLM, they also have limitations. For instance, short-answer QA does not capture partial correctness or nuanced factual misunderstandings. Moreover, chance-level performance or superficial cue-based heuristics might inflate apparent accuracy, thereby obscuring genuine hallucination tendencies.

TruthfulQA (Lin et al., 2022) is a more specialized benchmark designed to assess how truthfully models respond across domains (e.g., health, finance, politics, and others). This resource contains 817 adversarial queries that reflect common misconceptions. In order to succeed, an LLM must avoid reproducing erroneous folk wisdom. Human annotators curated the questions and associated correct versus incorrect replies, so the benchmark directly measures a model's capacity to resist hallucinating well-known myths. In one configuration, the model must select the correct answer from multiple-choice options; in another, it must produce generative responses without straying into imitative falsehoods. While its coverage remains confined to English and specific categories of misconceptions, TruthfulQA is influential in demonstrating how models might either faithfully present real facts or drift into widely circulated untruths.

Another recent benchmark is DefAn (Definitive Answers), which contains a large-scale dataset of over 75,000 QA prompts (Ashikur Rahman et al., 2024). DefAn features a hidden private set and public data to enable unbiased evaluation of novel

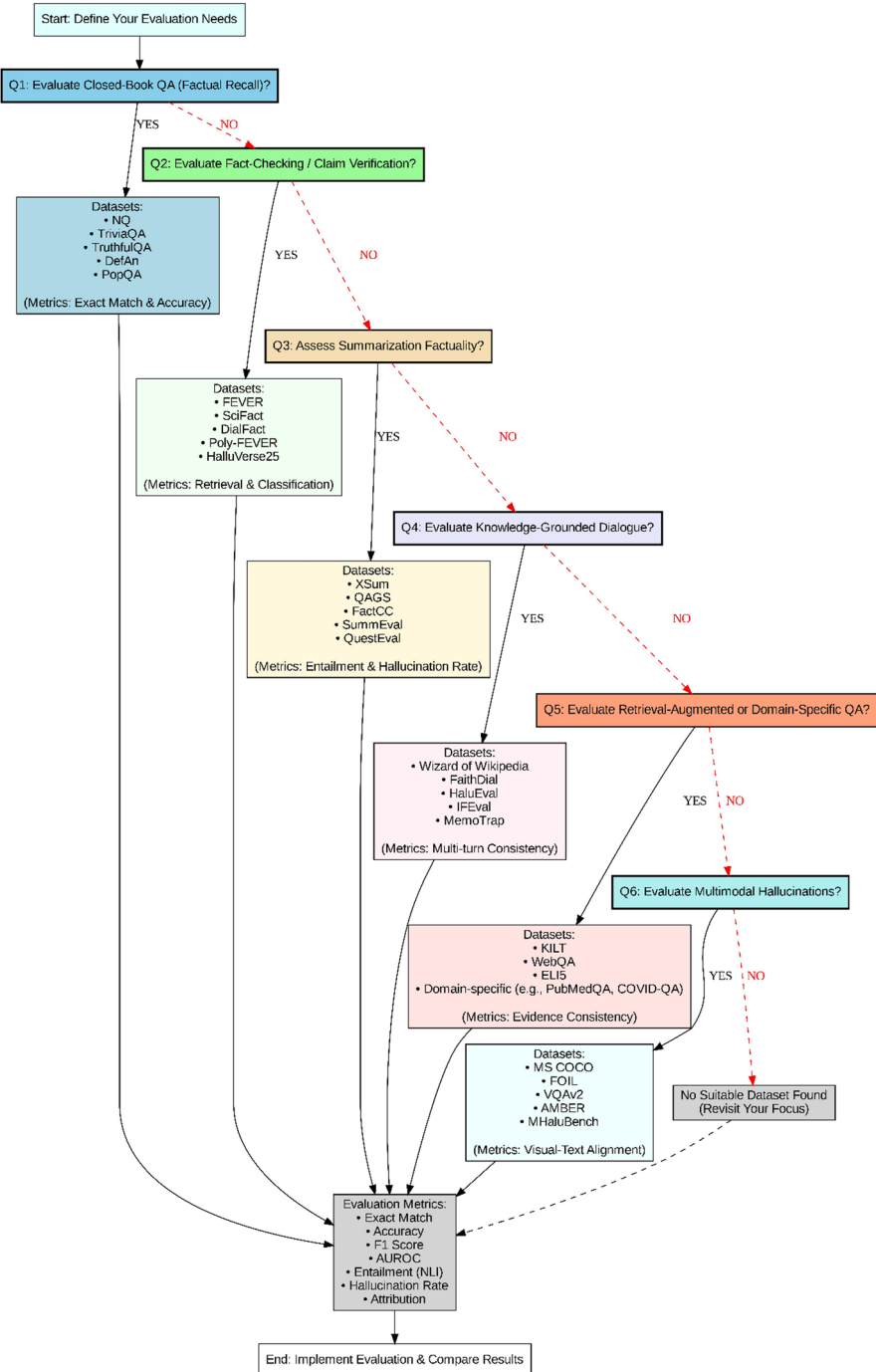


Fig. 5 Benchmark datasets and metrics

Table 3 Hallucination evaluation benchmarks

Category	Benchmark	Description	Size	Source/content	Question/ Task types	Evaluation metrics	Hallucination Types/Indication	Strengths	Limitations	Domain Focus	Multi- lingual	Features
Factual QA	Natural Questions (NQ)	Collection of real user queries	Large-scale	Real user queries	Factual, entity-based	Exact-match accuracy	Incorrect details presented as facts	Reflects real user information needs	Short-answer format doesn't capture nuance	General	No	Based on Google search queries
	TriviaQA	Trivia-style questions with factual answers	Large-scale	Trivia contests	Entity attributes, names, dates	Exact-match accuracy	Substantial deviation from verified ground truth	Comprehensive coverage of general knowledge	May not capture partial correctness	General	No	Includes both web and Wikipedia evidence
	TruthfulQA	Adversarial queries reflecting common misconceptions	817 questions	Human-crafted	Health, finance, politics, etc	Multiple-choice accuracy, generative correctness	Reproduction of erroneous folk wisdom	Directly measures resistance to common myths	Limited to English and specific misconception categories	Multiple (health, finance, politics)	No	Specifically designed to test truthfulness
DefAn	DefAn	Large-scale dataset with definitive answers	75,000 + QA prompts	Automated generation	Domain-specific with numeric elements	Factual accuracy	Straying from established ground truths	Hidden private unbiased evaluation, leaderboard support	Potential biases in automated prompt generation	Multiple	No	Public and private test sets
	PopQA	Knowledge-probing tasks with fill-in-the-blank format	Not specified	Knowledge probes	Factoid prompts	Completion accuracy	Incorrect completions or invented facts	Diagnoses knowledge gaps	Indirect measure of hallucination	General	No	Focused on popular entities

Table 3 (continued)

Category	Benchmark	Description	Size	Source/content	Question/Task types	Evaluation metrics	Hallucination Types/Indication	Strengths	Limitations	Domain Focus	Multilingual	Features
Fact-checking	FEVER	Fact Extraction and Verification	~184,000 claims	Wikipedia	Claim verification	Retrieval + classification accuracy	Incorrect classification of claims	Structured approach to fact-checking	Simple/artificial claims, single-sentence evidence	General	No	Supported, refuted, not enough information categories
	SciFact	Scientific claim verification	Not specified	Peer-reviewed papers	Scientific claim verification	Abstract-based verification accuracy	Technical misinformation	Domain-specific technical knowledge evaluation	Limited to scientific domain	Scientific	No	Uses scientific abstracts as evidence
	DialFact	Dialogue-based fact verification	Not specified	Conversational exchanges	Verification during interactive change	Wikipedia-based verification	Conversational falsehoods	Reflects real conversational hallucinations	Assumes Wikipedia covers ground truth	Conversational	No	Evaluates facts in dialogue context
	Poly-FEVER	Multilingual extension of FEVER	Not specified	Wikipedia (multiple languages)	Cross-lingual claim verification	Same as FEVER	Cross-language discrepancies	Addresses hallucinations across languages	Relies on translated content	General	Yes	Extends FEVER to multiple languages
	HalluVerse25	Fine-grained multilingual hallucination detection	Not specified	Multiple sources	Entity, relation, sentence-level verification	Fine-grained categorization	Granular detection at entity/relation level	Detailed analysis of hallucination types	Some synthetic perturbations may not reflect natural hallucinations	General	Yes (25 languages)	Fine-grained hallucination categorization

Table 3 (continued)

Category	Benchmark	Description	Size	Source/content	Question/ Task types	Evaluation metrics	Hallucination Types/Indication	Strengths	Limitations	Domain Focus	Multi- lingual	Features
Summarization	XSum	BBC articles with single sentence summaries	Not specified	BBC news	Single-sentence summarization	Factual consistency	Content not in source	Reveals high propensity for hallucinations	News domain bias	News	No	Single-sentence summary format
	QAGS	Question Answering for Generative Summaries	Not specified	CNN/DM, XSum	QA-based evaluation	QA accuracy	Inconsistencies between source and summary	Measures specific factual discrepancies	May miss subtle inconsistencies	News	No	Generates questions from summaries
FactCC	Factual Consistency Checker		Not specified	CNN/DM, XSum	Entailment classification	Classifier accuracy	Unsupported content	Trains classifier to identify hallucinations	May penalize correct but unsourced information	News	No	Synthetic+human labels approach
	SummEval	Meta-evaluation of summarization systems	Not specified	CNN/DM, XSum	Multiple system outputs comparison	Correlation with human judgments	Factual inconsistencies	Large-scale comparison across models	News domain focus	News	No	Meta-evaluation of multiple metrics
QuestEval	QA-based evaluation strategy		Not specified	Various	Question generation from source	Answer comparison	Discrepancies between source and summary answers	Detailed evaluation of factual alignment	Not a standard alone dataset	Multiple	No	Question generation for evaluation

Table 3 (continued)

Category	Benchmark	Description	Size	Source/content	Question/ Task types	Evaluation metrics	Hallucination Types/Indication	Strengths	Limitations	Domain Focus	Multi- lingual	Features
Knowledge-grounded Dialogue	Wizard of Wikipedia (WoW)	Dialogue with Wikipedia references	Not specified	Wikipedia passages	Knowledge-based conversation	Knowledge F1 (lexical overlap)	Fabricated facts beyond reference	Explicit knowledge grounding	Mainstream topics focus	Conversational	No	Turn-based dialogue structure
	FaithDial	Corrected version of WoW	Not specified	Wikipedia	Manual correction of hallucinations	Manual correction	Explicitly labeled faithful/unfaithful	Human-edited hallucinations	Retains WoW domain limitations	Conversational	No	Human corrections of hallucinations
	HaluEval	Dialogue component of hallucination leaderboard	Multiple sources	Various	Human annotation of factual errors	Annotation accuracy	Annotated factual errors	Diverse conversational prompts	Potential overlap with other benchmarks	Conversational	No	Part of HuggingFace leaderboard
	IFEval	Instruction-following evaluation	Not specified	Explicit user instructions	Adherence to instructions	Completion rate	Deviation from instructions	Measures instruction compliance	Not classic factual hallucination	Instructional	No	Tests instruction following
	MemoTrap	Tests memorization vs. instruction following	Not specified	Explicit constraints	Adherence to constraints	Constraint violation rate	Memorized but incorrect content	Highlights pattern recall issues	Indirect measure of hallucination	Instructional	No	Deliberate memory traps

Table 3 (continued)

Category	Benchmark	Description	Size	Source/content	Question/ Task types	Evaluation metrics	Hallucination Types/Indication	Strengths	Limitations	Domain Focus	Multi- lingual	Features
Retrieval-augmented Retrieval-augmented	KILT	Knowledge-Intensive Language Tasks	Not specified	Wikipedia	Multiple knowledge tasks	Answer correctness + relevance	Incorrect answers or irrelevant retrievals	Unified retrieval framework	Wikipedia-only limitation	General	No	Common retrieval framework
	WebQA	Web-based question answering	Not specified	Web content	Comprehensive explanations	Claim alignment with retrieved text	Ungrounded statements	Multi-source evidence	Complex retrieval challenges	General	No	Web-based evidence retrieval
	ELI5	Explain Like I'm Five	Not specified	Multiple sources	Multi-sentence explanations	Evidence support for claims	Statements without evidence	Long-form explanations	Retrieval failures risk	General	No	Simple explanations of complex topics
	PubMedQA	Medical domain main Q&A	Not specified	PubMed articles	Bio-medical questions	F1 performance	Domain-specific fabrications	Specialized knowledge	Small dataset size	Medical	No	Based on biomedical literature
	COVID-QA	COVID-19 specific information	Not specified	COVID research	Pandemic-related queries	Factual accuracy	Medical misinformation	Timely medical knowledge	Narrow domain focus	Medical	No	Focused on pandemic information
	MedHallu	Medical hallucination detection	Not specified	Medical literature	Plausible incorrect details	Detection accuracy	Plausible medical falsehoods	Tests specific medical knowledge	Limited public availability	Medical	No	Detects plausible medical falsehoods
	SelfCheck-GPT	Self-consistency evaluation	Not specified	Multiple model responses	Consistency checking	Contradiction detection	Internal contradictions	No external reference needed	Consistency ≠ correctness	Multiple	No	Uses model's own responses

Table 3 (continued)

Category	Benchmark	Description	Size	Source/content	Question/ Task types	Evaluation metrics	Hallucination Types/Indication	Strengths	Limitations	Domain Focus	Multi- lingual	Features
Multimodal	MS COCO Captions	Image captioning dataset	Not specified	Images+captions	Caption- ing	BLEU, object halluci- nation rate	Non-existent entities	Standard bench- mark	Basic overlap metrics	Visual	No	Multiple captions per image
	FOIL	Find One Mismatch in Image Caption	Not specified	Images+captions	Mis- match detection	Identi- fication accuracy	Single false detail	Targeted diagnostic	Limited to single errors	Visual	No	Detects single false entities
	VQAv2	Visual Question Answering	Not specified	Images+questions	Short answer generation	Answer accuracy	Unobservable content	Question diversity	Limited to short answers	Visual	No	Questions about visual content
	AMBER	Multimodal hallucination evaluation	Not specified	Images+text	Multiple tasks	Various	Image-inconsis- tent content	Standard- ized evaluation	Recent benchmark (limited usage)	Visual	No	Compre- hensive multimodal evaluation
	MHaluBench	Multimodal Hallucination Benchmark	Not specified	Images+text	Free- form generation, object detection	Consis- tency checks	Various visual hallucinations	Compre- hensive evaluation	Recent benchmark (limited usage)	Visual	No	Multiple visual hal- lucination types

Table 4 Evaluation metrics used across benchmarks

Metric	Description	Application context	Advantages	Limitations	Hallucination indication	Mathematical basis
Exact-match (EM)	Binary measure of answer correctness	QA with canonical answers	Simple, clear	Overly strict, misses partial correctness	Any deviation from ground truth	Binary match
Accuracy	Proportion of correct answers	Classification problems	Intuitive	May not reflect nuance	Incorrect classification	Correct/Total
Precision (P)	Proportion of true positives (TP) among positive predictions	Hallucination detection	Measures false alarm rate	Doesn't account for missed hallucinations	False positive (FP) detection	TP/(TP+FP)
Recall (R)	Proportion of true positives identified	Hallucination detection	Measures detection completeness	Doesn't penalize false positives	Missed hallucinations	TP/(TP+FN)
F1 Score	Harmonic mean of precision and recall	Balanced detection	Combined measure	Single metric obscures trade-offs	Balance of detection and precision	$2 * (P * R)/(P + R)$
AUROC	Area Under Receiver Operating Characteristic	Confidence ranking	Measures ranking quality	Insensitive to class imbalance	Confidence vs. correctness	Integral of ROC curve
Entailment	Whether source supports generated text	NLI-based evaluation	Semantic understanding	Relies on NLI model quality	Non-entailed content	Model-dependent
Hallucination rate	Percentage of outputs with factual errors	Overall assessment	Simple summary statistic	Binary, misses severity	Presence of any error	Error count/Total
Attribution/ Grounding	Fraction of output traceable to source	RAG evaluation	Direct source connection	May miss implicit connections	Unsupported segments	Supported content/Total
Knowledge F1	Lexical overlap with reference knowledge	Dialogue grounding	Simple implementation	Surface-level matching	Low overlap with source	F1 on knowledge terms
BLEU	Bilingual Evaluation Understudy	Image captioning, text generation	Industry standard for generation tasks	Surface-level n-gram matching	Low overlap with reference	Precision of n-grams with brevity penalty
Object hallucination rate	Proportion of captions with nonexistent entities	Image captioning	Directly measures visual hallucinations	Binary measurement	Generated entities not present in image	Nonexistent entities/Total captions

Table 4 (continued)

Metric	Description	Application context	Advantages	Limitations	Hallucination indication	Mathematical basis
FOIL detection accuracy	Ability to identify mismatched words in captions	Visual-textual verification	Targeted detection of errors	Limited to single mismatch scenarios	Failure to identify false details	Correct identifications/Total examples
VQA accuracy	Correctness of answers to visual questions	Visual Question Answering	Measures visual understanding	Partial credit often not given	Content not observable in image	Agreement with human answers
Free-form generation consistency	Semantic alignment between image and generated text	Multimodal generation	Captures nuanced hallucinations	Difficult to automate evaluation	Semantically inconsistent content	Often requires human evaluation
Object detection precision	Accuracy of detected objects in image-based responses	Vision-language tasks	Grounded in visual content	Doesn't capture relationships	Mentioning objects not detected	Correctly detected objects/Total mentioned
Consistency checks	Agreement between different aspects of multimodal output	Cross-modal verification	Detects internal contradictions	Complex implementation	Cross-modal inconsistencies	Task-dependent measures

content and to support a leaderboard system. This benchmark emphasizes queries with definitive answers; hence, it is well-suited for detecting factual hallucinations when a model strays from established ground truths. Empirical reports indicate that across various large models, factual hallucination rates on DefAn's public portion range from around 59% to over 80%, which suggests that domain-specific queries with numeric elements or other specifics pose heightened challenges. Because it is newly introduced, questions remain about potential biases in automated prompt generation, but its breadth and domain diversity make it valuable for consistent cross-model comparisons.

Knowledge-probing tasks such as PopQA (Mallen et al., 2023) can also indirectly diagnose hallucinations. These tasks typically involve asking the model to fill in blanks in factual statements (for instance, "Paris is the capital of ___") or to respond to short factoid prompts. While such resources do not always include a direct measure of hallucination, incorrect completions or invented facts signal potential weaknesses in factual reliability.

6.2 Fact-checking and claim verification datasets

A second major class of benchmarks approaches hallucination detection as a fact verification or claim-checking problem. These datasets typically contain declarative statements labeled as supported, refuted, or unverified relative to trusted sources.

Models that fail to accurately classify or justify claims often do so because they introduce hallucinated content that lacks support in the evidence. One such dataset is the Fact Extraction and VERification (FEVER) (Bekoulis et al., 2021). This benchmark contains approximately 184,000 claims derived from Wikipedia, each labeled as supported, refuted, or insufficient evidence based on whether Wikipedia evidence can justify or contradict the statement. Evaluation typically involves both retrieval (i.e., locating relevant sentences) and classification (labeling each claim). While FEVER's structured approach benefits experiments in fact-checking, many claims are simple or artificially constructed, which can be less representative of the free-form text generated by LLMs. In addition, single-sentence evidence references may not suffice for detecting more complex or context-dependent forms of hallucination.

Specialized counterparts such as SciFact (Wadden et al., 2020) focus on scientific claims drawn from peer-reviewed papers. When evaluated on this benchmark, GAI models must decide whether a claim is supported or refuted by the abstract of a relevant article, demonstrating how confidently or erroneously an LLM might handle technical or domain-specific knowledge. DialFact addresses similar concerns in dialogue settings, requiring verification of statements uttered in conversation (Gupta et al., 2021). Here, the claim to be verified occurs naturally during the interactive exchange, which makes it more reflective of real conversational hallucinatory behaviors. One limitation is that DialFact still assumes that Wikipedia covers the ground truth for each claim; however, in highly specialized/niche discussions, verifying claims can be more challenging if the knowledge source is incomplete.

In order to address hallucinations in multiple languages, Poly-FEVER (Zhang et al., 2025a) extends FEVER-style tasks to cover non-English statements, and HaluVerse25 (Abdaljalil et al., 2025) further introduces fine-grained annotations that categorize hallucinations at the entity, relation, or sentence level across multiple languages. Such multilingual benchmarks can be essential for investigating whether LLMs hallucinate differently across languages, particularly when training data or domain knowledge are scarce. One caveat is that some of these datasets rely on synthetic perturbations of factual statements, raising concerns that models might learn to detect patterns specific to artificially generated examples rather than learn robust factual fidelity.

6.3 Summarization factuality benchmarks

As discussed earlier, abstractive summarization is a domain where hallucinations frequently occur. Thus, several specialized benchmarks seek to measure how accurately summarization models capture the source content. Early observations on the XSum dataset (Narayan et al., 2018) (BBC articles paired with single-sentence summaries) revealed a high propensity for models to hallucinate. Researchers responded by creating specific subsets and annotation layers, such as QAGS (Questions Answering for Generative Summaries) (Wang et al., 2020) and FactCC (Iwamoto & Shimada, 2024), in which human annotators label each model-generated summary as either faithful or containing factual errors. FactCC uses a synthetic-and-human-labeled set of summaries from CNN/DM and XSum, then trains an entailment classifier to identify unsupported content. This approach has informed a broader line of research that

uses natural language inference (NLI) to gauge factual consistency between a source document and a proposed summary. Although these benchmarks provide rigorous ways to categorize and measure hallucinations, they sometimes penalize a model for introducing additional but correct context that is not explicitly present in the source. As a result, “unfaithful but factually true” content can be labeled as a hallucination to underline the difference between local faithfulness (i.e., a summary must only contain details derivable from the source) versus global factual correctness (i.e., the statement might be correct but not traceable to the source text).

It is worth noting that datasets such as SummEval (Fabbri et al., 2021) aggregate multiple system outputs to provide a large-scale meta-evaluation of the correlation between different metrics and human judgments of factual consistency. Meanwhile, recent benchmarks [e.g., QuestEval (Shen et al., 2023)] leverage QA-based strategies to generate questions from the source document and compare the summary’s answers with those in the original text. Although such benchmarks do not constitute datasets in themselves, they rely on QA pairs for implementation and are often validated against annotated sets (such as XSum-Fact (Shen et al., 2023)). The news domain bias in these summarization resources is a potential limitation; long-form or domain-specific documents may exhibit different hallucinations. Furthermore, high-scoring outputs might merely copy extensive portions of the source text verbatim, potentially obscuring deeper challenges of generating concise yet accurate summaries.

6.4 Knowledge-grounded dialogue benchmarks

Hallucinations in dialogue-based systems typically manifest as fabricated facts delivered in a conversational style. To study this phenomenon, knowledge-grounded dialogue benchmarks provide specific reference materials and require participants to incorporate those facts into conversational replies without inventing unsupported details.

For example, Wizard of Wikipedia (WoW) (Dinan et al., 2018) is an early example of this class of benchmarks. Relevant Wikipedia passages accompany the dialogue; the system is expected to produce responses explicitly grounded in that knowledge. Standard evaluation strategies include computing Knowledge F1 (Zhang et al., 2021), which measures lexical overlap between the model’s reply and the reference wiki text. Although WoW is valuable for training and evaluating grounded conversational performance, this benchmark focuses on relatively mainstream topics and does not thoroughly address multi-turn consistency or more subtle forms of hallucination that extend across several dialogue turns.

FaithDial (Dziri et al., 2022b) refines the Wizard of Wikipedia by manually correcting or rewriting hallucinated responses in WoW dialogues, with human annotators systematically editing any utterance that introduced unsupported facts. This led to the creation of a new dataset in which each turn is explicitly labeled as faithful or not. The same allows researchers to train discriminators that detect hallucinated statements and generative models that learn to remain faithful to the provided text. The major limitation is that it retains WoW’s domain characteristics, so it may not capture the full breadth of real-world dialogue contexts. However, it is a rare exam-

ple of a dataset in which human raters have meticulously identified and rewritten hallucinations.

The HuggingFace Hallucination Leaderboard (Hong et al., 2024) also includes a dialogue component (HaluEval), in which a range of model outputs has been annotated for factual errors. This aggregated resource provides a pure evaluation subset without training. Because it is drawn from multiple sources, HaluEval can reflect a more diverse set of conversational prompts. Nevertheless, some of the data may overlap with FaithDial or other knowledge-grounded benchmarks, and the documentation for each portion is sometimes limited.

Instruction-following dialogues represent yet another variant. For instance, the IFEval (Instruction Following Evaluation) dataset and its variants (Kovalevskyi, 2024), as well as MemoTrap (Shi et al., 2024), highlight scenarios in which a model must adhere to explicit user instructions that might forbid or constrain certain responses. Hallucination in these benchmarks emerges when the model deviates from the request by introducing memorized but incorrect content. While this may not always represent a classical factual hallucination, it shows how LLMs often rely on spurious pattern recall at the expense of faithfully following instructions.

6.5 Retrieval-augmented and knowledge-intensive benchmarks

Retrieval augmentation offers another strategy for addressing hallucinations by enabling models to consult external sources. For example, the KILT (Petroni et al., 2021) benchmark suite integrates multiple knowledge-intensive tasks within a common retrieval framework. Models are penalized for producing incorrect answers and citing irrelevant or incorrect retrievals since genuine grounding requires correct evidence. This unified setting encourages RAG development that reduces hallucinations through explicit evidence checking. However, KILT's exclusive use of Wikipedia may not generalize to real-world contexts where knowledge is distributed across the web, enterprise repositories, or private databases. Other open QA resources, such as WebQA (Chang et al., 2022) and ELI5 (Fan et al., 2019), ask models to produce comprehensive, often multi-sentence explanations. Since these tasks may involve drawing on numerous fragments of textual evidence, the risk of hallucination increases if retrieval or evidence selection fails. Scoring can be based on how many unique claims in a generated answer align with the retrieved text, reflecting the degree to which the model grounds its statements in verifiable information.

Then, domain-specific QA benchmarks illustrate how hallucinations can carry particularly severe risks in specialized domains. For example, PubMedQA (Jin et al., 2019) and COVID-QA (Möller et al., 2020) evaluate scientific or biomedical questions; smaller curated sets, such as MedHallu (Pandit et al., 2025), inject plausible but incorrect details to assess whether an LLM can detect or avoid them. Similar proprietary datasets exist for law (Bommarito et al., 2021; Sovrano et al., 2021). Another dimension involves self-consistency checks or multi-step reasoning. Methods like SelfCheckGPT (Manakul et al., 2023) prompt the model to produce multiple responses to the same prompt then evaluate internal contradictions. Although consistency does not necessarily guarantee correctness, finding contradictory outputs

can be a useful proxy for spotting potential hallucinations when factual claims vary across repeated attempts.

6.6 Multimodal and image generation benchmarks

Hallucinations also arise in multimodal systems, including image captioning, visual question answering (VQA), and text-to-image generation. Such Large Vision-Language Models (LVLMs) risk generating textual descriptions that mention objects or properties not present in the image or might fail to describe key elements. For example, MS COCO (Lin et al., 2014) captions were an early standard for image captioning. Visual question answering tasks, such as VQAv2 (Goyal et al., 2019), also indirectly measure hallucinations. The model receives an image-question pair and must provide a correct short answer. If the predicted answer includes content that is not observable in the image, that content is, by definition, hallucinated. Because VQA questions can be more open-ended, large vision-language models sometimes provide plausible details that are not truly grounded in the image. Newer benchmarks [e.g., AMBER (Wang et al., 2023c), MHALuBench (Chen et al., 2024c), etc.] are attempting to standardize the evaluation of vision-language models by incorporating free-form generation, object detection, consistency checks, and other methods to measure the severity and nature of image-based hallucinations.

6.7 Common evaluation metrics

Across the listed benchmarks, several quantitative indicators consistently appear (see Table 4). The simplest measures, exact-match (EM) and accuracy, are used primarily in QA contexts with canonical correct answers. Models are scored by the proportion of their answers that match ground-truth responses, possibly after string normalization. Although convenient, these metrics can be overly strict, failing to reward partially correct or more nuanced factual statements. When hallucination detection is framed as a classification problem, precision, recall, and their harmonic mean (F1) often serve as the main metrics. Some studies also report the area under the receiver operating characteristic curve (AUROC) to assess how well a detection model ranks hallucinated outputs above correct ones in terms of confidence. In practice, precision-recall trade-offs matter: detection with high recall might flag most hallucinations but risk incorrectly labeling many true statements.

Other standard metrics, such as BLEU (Papineni et al., 2002), measure overlap with reference captions, more targeted evaluations examine whether the system mentions entities that do not appear in the image. Studies compute an object hallucination rate, defined as the proportion of generated captions that include at least one nonexistent entity. Find One Mismatch in the Image Caption (FOIL) (Shekhar et al., 2017) presents a related diagnostic: given an image and two captions (one containing a single false detail), the model must identify the accurate caption or pinpoint the mismatched word. Entailment-based approaches lean on NLI models to judge whether the source supports a generated text. Some extend this approach to the next level by extracting factual tuples from both the source and the generation, and then measuring overlap or contradiction. A high overlap (or precision with respect to the source's fac-

tual graph) indicates few hallucinated elements. Another single “hallucination rate” can be defined as the percentage of outputs containing at least one factual error. Then, attribution or grounding metrics measure the fraction of the model’s output traceable to a retrieved source. In retrieval-augmented pipelines, the reference text must fully justify the answer; otherwise, any unsupported segments represent hallucinations.

7 Case studies in hallucination impact and mitigation

While much of the discussion on hallucinations and their remedies focuses on theoretical frameworks and experimental outcomes, practical insights often surface when one examines real-world implementations (Lee et al., 2025; Zhang et al., 2025b). This section presents qualitative and general examples from three specific domains to illustrate how hallucinations can emerge, undermine trust, and how targeted strategies can mitigate their worst repercussions. A sample companion flowchart that could be helpful in other areas is seen in Fig. 6.

7.1 Academic research

When researchers experiment with generative models for tasks such as manuscript drafting or citation recommendations, the potential for invented references looms large. Thus, hallucinations in a scholarly setting can pose considerable risks to scientific integrity (Chakrabarty et al., 2024). In certain environments, early prototypes have been used to generate outlines of literature reviews (Pérez-Guerrero et al., 2024). However, it was discovered that when the system encountered incomplete or contradictory data, it sometimes stitched together plausible but inaccurate citations (Erdem et al., 2025). These spurious references (if left unchecked) could degrade the quality of subsequent analytical work and cause confusion among peers.

Some research groups have developed qualitative measures to address these problems by focusing on the collaborative role between the automated tool and the researcher. One representative strategy could be for the model to produce a preliminary text, and for human reviewers to examine each claim and trace the references back to established publications (Naser et al., 2024a). Rather than discarding generative output entirely, the reviewers selectively retain ideas that align with verifiable data (i.e., discarding sentences that appear inconsistent) (Jacob et al., 2025). When carefully managed, the synergy between human expertise and automated suggestions can accelerate tasks such as summarizing bodies of work, drafting methodological sections, or suggesting new angles for investigation (Gabriel et al., 2021). Another aspect that surfaces in academic usage relates to the distortion of specialized terminology. Occasionally, a model might skillfully mimic the scientific writing style but misapply domain-specific concepts, producing paragraphs that appear authoritative yet rest on misinterpretations of fundamental principles. Therefore, while experimenting with these tools, one can emphasize the importance of training users to detect subtle pitfalls.

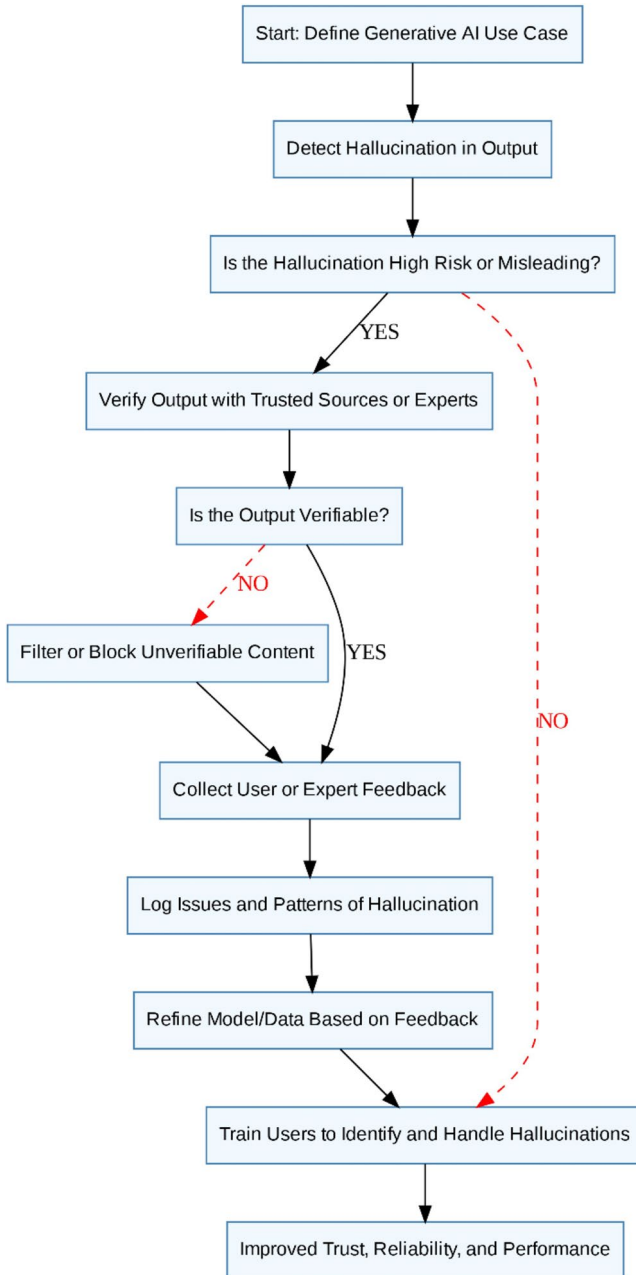


Fig. 6 Generic flowchart for possible examination of hallucinations

7.2 Commercial deployment

In commercial contexts, generative models are deployed for marketing, customer support, and content creation activities. The impact of hallucinations in these spheres varies depending on the target audience and the nature of the misstatements (Hicks et al., 2024). In certain consumer-facing applications, such as chatbots, minor factual lapses can annoy users but rarely cause deep harm (Hostetter et al., 2024; Naser et al., 2024b). On the other hand, an erroneous recommendation about product use or an incorrect claim about financial services can undermine consumer trust and provoke legal scrutiny (Jones, 2025). In such instances, companies generally respond by implementing multiple layers of oversight that combine automated scanning tools with a curated knowledge base of product details to ensure the system's responses remain aligned with corporate standards. In practice, corporations highly value reliability, as brand damage from repeated hallucinations can be difficult to reverse. Therefore, corporations frequently constrain outputs rather than allowing the model unrestricted freedom to form responses. This measure helps protect brand reputation by minimizing speculation. For example, a corporation might incorporate a mechanism that automatically blocks any generated answer that references non-existent merchandise. While such constraints limit the scope of permissible replies, they foster greater reliability.

An evolving trend is the incorporation of end-user feedback at scale (Kaate et al., 2025). Many customer support chatbots now request user validation in real-time. Therefore, if a client flags a response as misleading, the incident is logged in a monitoring system that aggregates similar incidents to identify recurring blind spots in the model. From there, specialized engineering teams refine the underlying data pipeline or adjust weighting factors that shape the model's behavior (Dinakar et al., 2025). Through this cyclical improvement process, it becomes possible to address sources of hallucination that might not surface during pre-deployment trials. Thus, large-scale user bases essentially collaborate in policing GAI's accuracy, albeit unintentionally.

7.3 Educational contexts

In education, generative models have begun assisting students with various tasks (e.g., test preparation) (Smith et al., 2025; Talha Junaid et al., 2024). The capacity of such systems to give immediate feedback, parse large volumes of text, or adaptively present tutorials holds considerable promise. Yet the challenges associated with hallucinations remain non-trivial, as in a classroom environment, inaccuracies propagated by GAI tools can distort a student's foundational understanding (Abdelnabi et al., 2024). Upon experimenting with generative platforms, some educators have encountered misleading accounts that appear valid but are not (Yusuf et al., 2024). This is especially problematic because younger learners may lack the critical thinking skills or domain knowledge needed to identify falsehoods. Furthermore, if teachers themselves rely heavily on these systems without thorough checks, the errors can perpetuate or even multiply throughout a cohort of learners.

Various tactics have taken shape to address some of the above risks in educational contexts (Zamfirescu-Pereira et al., 2025). For example, a number of schools

implement structured lesson plans in which students must corroborate AI-driven outputs with traditional research methods. Here, students are guided to apply external validation procedures, such as consulting library databases or group discussions with classmates (Liu et al., 2024b). This practice not only reduces the chance of absorbing incorrect facts but also develops digital literacy skills. In addition, others may experiment with curated GAI systems that restrict themselves to academic resources maintained by the school and filter out unverified websites or user-generated datasets to maintain a more consistent quality of information. While helpful, some view this practice as potentially limiting the breadth of available knowledge, undercutting the open-ended exploration that many educators value. Another dimension arises in how GAI tools might be used to foster critical thinking and provoke student inquiry (Jauhainen & Guerra, 2025; Naser, 2022). For instance, a teacher might invite the class to analyze a short GAI-generated text, identify potential mistakes, or question its assumptions. This exercise turns the risk of hallucination into a teachable moment to sharpen students' ability to evaluate sources, reflect on bias, and verify information.

8 Future research directions and limitations of this review

As seen in the above review, generative models can produce human-like output fluently, but their tendency to hallucinate remains an impediment to reliable adoption (Bommasani et al., 2021). This section presents some promising research ideas and concludes by outlining the limitations of the presented review.

8.1 Future research directions

As this field advances, numerous unsettled questions invite further examination. While present techniques demonstrate success in curbing explicit hallucinations, they have not entirely eradicated subtler forms of misrepresentation that arise when data is ambiguous. In many cases, generative systems lack a principled mechanism for signaling incomplete knowledge, potentially masking areas where additional research or contextual detail might be warranted. Future work could explore frameworks that allow models to gracefully indicate the boundaries of their expertise, thereby reducing the likelihood of filling those gaps with unsubstantiated claims.

Another area that merits attention concerns the unification of GAI with domain-specific knowledge (Lang et al., 2025; Sabbagh et al., 2024). Although retrieval from external sources has proven valuable, future research may concentrate on dynamic knowledge injection, in which generative systems continually update their internal representations based on newly published data or streams. Such a mechanism would require model architectures capable of reconciling parametric knowledge with regularly refreshed memory. In principle, this could minimize the risk of hallucinating outdated information while, at the same time, would also expand the capacity for long-form reasoning. However, implementing these dynamic knowledge modules at scale may pose a suite of technical barriers related to memory management and indexing efficiency, which determine which updates to include in the new version

of the GAI system. This particular theme may present various sub-areas for future investigation (Hatalis et al., 2023).

One should also recognize the need to avoid misstatements in multiple domains (as opposed to a particular domain). Here, transfer learning could serve as a potential solution for cross-domain hallucination detection (Fan et al., 2025). Through more targeted transfer learning, knowledge from stable domains can be leveraged to improve reliability in less-examined fields. This effort might involve modular architectures capable of transplanting validated domain knowledge onto expansions that handle new subject matter. In parallel, unsupervised hallucination detection represents another frontier, particularly for domains lacking robust labeled data. For example, models that autonomously detect anomalies in their own outputs could continually refine themselves without waiting for curated error samples.

Another promising direction involves bridging text-based reasoning with sophisticated data representations (e.g., multi-modal embeddings and structured domain ontologies) (Zhang et al., 2024b). Here, generative systems could anchor their outputs in a substrate less prone to ambiguous interpretations by combining textual input with numerical or graphical schemas. On the one hand, multi-modal training regimens that bind textual tokens to representations of tables, images, or graphs might reduce the model's inclination to fabricate details. On the other hand, data source combinations will likely require integrated training pipelines to align heterogeneous inputs without diluting the model's generative competence.

As GAI models grow in size and complexity, it becomes increasingly difficult to trace the specific chain of associations that leads to a hallucinated statement. Therefore, one possible means to examine the behavioral tendencies of generative systems is through the lens of interpretability (Saha et al., 2023; Tan et al., 2024). At a deeper level, the crucial step in this theme is developing robust mechanisms to verify whether an internal reasoning sequence is consistent with known references. For instance, future research might emphasize the design of narrative-based explanations, in which the model can self-report its reasoning chain in a structured format. Such synergy might foster a new family of GAI systems that actively clarify how they arrived at their conclusions, thereby reducing the chance of undetected hallucinations.

In GAI systems that rely on textual interaction, linguistic nuances can cause confusion, especially when a model with primarily English data responds to queries in less common languages (Guerreiro et al., 2023). Here, hallucinations may amplify when the training data is sparse or when certain cultural references are incorrectly generalized. Future directions may thus emphasize robust cross-lingual knowledge alignment to ensure that the model consistently applies factual anchors across language boundaries. This issue could be addressed through carefully designed collaborative efforts among computational linguists, domain experts, and local communities.

GAI models seem to perform well at learning distributional features, but do not systematically test their statements against any self-defined verification objective. This implies that a future line of inquiry may revolve around the potential for self-supervised correctness feedback (Kamoi et al., 2024). Implementing such an internal review mechanism could be meritorious, as this design could reform the way generative models handle uncertain prompts (though it may also raise questions about computational overhead). Another limitation in current research is the difficulty of

curating large volumes of labeled examples containing hallucinations. As models can hallucinate in unpredictable ways, this makes it hard to systematically gather negative samples that reflect the entire spectrum of plausible mistakes. Thus, another research direction could focus on methods for generating synthetic data to improve detection robustness (Johnson & Hajisharif, 2024).

Given that user preferences can vary dramatically across contexts, expertise, and the consequences of potential errors, quantifying user acceptance of minor inaccuracies or illusions raises interesting questions in human–computer interaction. So, another promising aspect pertains to user-driven customizations that adapt the aggressiveness of hallucination mitigation (Ha et al., 2024). At a broader level, developers of generative systems might experiment with bridging algorithmic advances to user-centric training programs. This reflects the notion that an informed user population serves as an additional safeguard that complements the measures discussed earlier. Future research might include designing integrated tutorials or built-in diagnostic modes that guide users to cross-check facts via external tools or community feedback.

8.2 Limitations of this review

This review seeks to synthesize the conceptual and methodological dimensions of hallucinations in GAI, but, in an unintended way, it restricts its inquiry primarily to theoretical and qualitative notions. In a way, the current undertaking focuses on thematic insights rather than extensive numerical assessments, which is hoped to be of interest to newcomers to this field and to complement the growing body of empirical work devoted to quantifying error rates, establishing performance benchmarks, and conducting large-scale user studies (Chelli et al., 2024; Wang et al., 2024). Nevertheless, interested readers are encouraged to consult complementary empirical research for a more detailed quantification of how frequently or severely hallucinations occur across particular model architectures.

Moreover, this review emphasizes the duality of detection and correction, recognizing that each encompasses a broad spectrum of techniques spanning natural language processing, computer vision, multimodal modeling, and human–computer interaction. Therefore, this review does not attempt an exhaustive survey of all techniques but rather aims to capture a representative range of current innovations. As a result, some recently introduced methods may not be covered in exhaustive detail, particularly if they are still in their early stages or lack sufficient documentation. An additional scope boundary concerns the rapid evolution of GAI, in which techniques for reducing hallucinations rely on temporary solutions that become less relevant as new training processes, architectures, or datasets emerge.

Given the high velocity of developments in this field, any static review faces the risk of partial obsolescence. Therefore, readers are advised to approach this review as a snapshot of current trends. Nevertheless, the review has been deliberately structured around framework-level contributions—a multi-dimensional taxonomy, comparative analyses of detection and correction paradigms, and a consolidated benchmark mapping—that are intended to retain organizational utility even as the specific models, datasets, and toolchains they classify evolve or are superseded. In this sense, the taxo-

nostic and comparative scaffolding offered here is designed to accommodate, rather than be invalidated by, the continued pace of innovation in generative AI. It must be stressed that an examination of hallucinations must be situated within an ongoing dialogue about the fluidity between correct, creative, and incorrect outputs (Emsley, 2023). Therefore, readers should be mindful that not all generative use cases demand strictly verifiable content.

9 Conclusions

Despite the continued advances in GAI and LLMs, hallucinations remain a significant barrier to their deployment. This paper reviews hallucinations through this lens and explains how they stem from intricate interactions among model architectures, training data characteristics, and inference strategies. In addition, various detection methods, ranging from post-hoc verification against knowledge bases to real-time uncertainty estimation, offer valuable tools for identifying fabricated content, each with its own set of strengths and limitations in terms of computational overhead and effectiveness across different error types. Similarly, a range of correction and mitigation strategies, from proactive training refinements to reactive hybrid systems, was reviewed. Other key findings include:

- A taxonomy for hallucinations can be classified along several dimensions, including intrinsic vs. extrinsic, factual vs. logical, domain specificity, scope, intentional vs. unintended, detection difficulty, and temporality.
- After production, post-hoc detection techniques analyze generated outputs using consistency checks, external fact-checking tools, contrastive learning, and specialized verification models.
- Uncertainty calibration methods estimate a model's confidence during generation to flag potentially erroneous outputs, using Bayesian approaches, self-consistency checks, and token-level uncertainty estimation.
- Correction and mitigation strategies can be proactive, intervening during training with specialized datasets and feedback loops, or hybrid, combining detection and correction elements.
- A diverse range of benchmark tests and datasets, including factual question-answering, fact-checking, summarization, and knowledge-grounded dialogue benchmarks, are used to evaluate hallucinations.
- Common evaluation metrics include exact match, accuracy, and hallucination rate. Each metric offers a different perspective on factual fidelity.

10 Glossary

This glossary defines key concepts and specialized terminology related to hallucinations in generative models.

10.1 Generative architecture

This refers to any framework designed to produce novel text, images, or other data modalities. Common structures include transformer-based language models and diffusion networks for image synthesis. Generative architectures typically learn probability distributions from large training data, allowing them to create outputs that mimic real-world patterns.

10.2 Hallucination

A phenomenon in which a generative model produces content that appears coherent but lacks factual or logical basis. Hallucinations may result from insufficient training data coverage, spurious correlations in the model parameters, or sampling methods that overemphasize rare tokens.

10.3 Parametric knowledge

Information encoded within the parameters of a model, as opposed to being explicitly accessible in an external data store. When a system relies heavily on parametric knowledge, it may yield outdated or mistaken content if the training data is incomplete or inaccurate.

10.4 Retrieval-augmented generation (RAG)

A technique that combines a generative engine with external databases or search systems. During generation, the model consults retrieved documents to anchor its responses in verifiable facts, thereby reducing speculation.

10.5 Uncertainty calibration

A family of techniques aimed at measuring how confident a model is in its own predictions. These methods can signal instances where the system is likely to produce errors to facilitate early interception or fallback procedures.

10.6 Post-hoc detection

An approach that evaluates generated outputs after they are fully formed. Fact-checking systems, style analyzers, and specialized classifiers can flag suspicious passages to force the model (or a secondary process) to verify or correct them.

10.7 Adversarial prompting

The practice of providing inputs designed specifically to expose vulnerabilities or misinterpretations in a generative model. Often used in *red-teaming* exercises, adversarial prompting helps identify scenarios that provoke hallucinations.

10.8 Confidence thresholding

A mechanism that sets a limit below which the model's outputs are either rejected or subjected to additional validation. When the predicted likelihood of correctness is too low, the system can decline to respond or request disambiguation.

10.9 Domain-specific data

A curated dataset containing specialized knowledge. Models trained on specific domain-related data often exhibit greater factual precision in niche domains, though they may lose the breadth of general knowledge.

10.10 Self-consistency

An internal property of a model's output that ensures the overall text or image does not contradict itself. Low self-consistency indicates a strong chance of misinformation, as repeated ideas or details can clash with prior statements.

10.11 Ground-truth dataset

A reference collection of accurate information is used to judge whether a model's outputs are factually aligned. Ground-truth datasets are especially vital in testing detection and correction strategies designed to reduce hallucinations.

10.12 Interactive verification

A workflow in which the model and the user engage in a dialogue to ascertain the correctness of particular claims. The system might prompt the user for clarifications or provide direct evidence supporting its assertions.

10.13 Multi-modal fusion

A capability wherein the model processes and generates more than one data type (e.g., text combined with images). Multi-modal architectures introduce complex risks of cross-modal hallucinations if each data stream is not accurately aligned.

10.14 Blackbox model

A system whose internal parameters or decision processes are opaque.

10.15 Interpretability

The extent to which observers can understand or explain a model's inner workings. High interpretability aids debugging efforts and fosters confidence that the system's outputs stem from coherent patterns rather than random guesses.

10.16 Tokenization

A fundamental step in language processing wherein text is split into discrete tokens, such as words or subwords. In some generative models, suboptimal tokenization can introduce confusion, increasing the likelihood of hallucinating content.

10.17 Synthetic benchmark

A controlled dataset or test suite manufactured to assess model performance on specific tasks. Synthetic benchmarks often incorporate intentionally misleading examples to evaluate how effectively a system identifies and corrects inaccuracies.

10.18 Regulatory sandbox

A controlled environment where new AI tools can be tested without exposing the public to unforeseen risks. In the context of hallucination mitigation, a sandbox provides a venue to evaluate the efficacy of novel interventions before broad deployment.

10.19 Transparency portal

An interface or platform provided by model developers to disclose metadata, training protocols, or references that underpin an AI system's outputs. Such portals have gained prominence as a means to mitigate concerns over unexplained hallucinations.

10.20 Human-in-the-Loop

A collaborative framework that enlists human experts to oversee or refine algorithmic outputs. The system mitigates the risk of presenting unchecked hallucinatory content by incorporating human oversight, particularly for high-stakes tasks.

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